
THE SHAPE OF ATTRACTION IN UMAP: EXPLORING THE EMBEDDING FORCES IN DIMENSIONALITY REDUCTION

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ABSTRACT

Uniform manifold approximation and projection (UMAP) is a widely used member of the broader family of neighbor embedding methods. Like many related dimensionality reduction approaches, it constructs a low-dimensional representation through attractive and repulsive interactions among high-dimensional data points. We analyze these interactions as functional shapes, showing how they influence cluster formation, visualization, and algorithmic behavior across UMAP and closely related methods. Repulsion largely accentuates differences by shaping cluster boundaries and inter-cluster distances. Attraction has a broader and less obvious role: depending on the current separation, nominally attractive updates can contract or expand distances in the embedding. This view explains why learning-rate annealing is important in UMAP and why attraction and repulsion are treated differently across methods. By modifying the attraction shape, we also improve the consistency of cluster formation under random initialization. Overall, the analysis provides a general lens for interpreting, stabilizing, and tuning UMAP-like embeddings without changing the datasets, tasks, or evaluation conditions considered here.

1 Introduction

Modern scientific and industrial workflows routinely produce high-dimensional datasets whose structure must be inspected without assuming a small set of predefined variables. Dimensionality reduction (DR) methods address this need by mapping such data into visual spaces where neighborhood, cluster, and separation patterns can be explored. Widely used approaches such as t -distributed stochastic neighbor embedding (Maaten and Hinton [2008]) and uniform manifold approximation and projection (UMAP) (McInnes et al. [2018]) share an attraction-repulsion mechanism: similar data points are drawn together while dissimilar points are pushed apart. Because these methods are unsupervised, their relevance extends beyond any single labeled benchmark. They infer and preserve structure from local attractive relationships and broader repulsive relationships, making them useful in settings such as clustering (Becht et al. [2019]), exploratory data analysis (Fleischer and Islam [2020]), anomaly detection in semiconductor manufacturing (Fan et al. [2021]), visual search (González-Márquez et al. [2024]), time series analysis (Altin and Cakir [2024]), studying representation convergence (Huh et al. [2024]), and outlier image detection (Islam and Fleischer [2024]). Across these applications, the common goal is not merely to draw a picture, but to expose otherwise hidden patterns in unlabeled or weakly labeled data. In this sense, DR illustrates how label-free optimization can turn complex data geometry into interpretable visual structure.

A growing collection of methods builds on the same attraction-repulsion template (Amid and Warmuth [2019], Agrawal et al. [2021], Wang et al. [2021], Narayan et al. [2021], Yang et al. [2022], Wang et al. [2025], Kury et al. [2025]), with each variant emphasizing different aspects of the data geometry. Although these algorithms are now applied across many domains, their practical use still depends on heuristics whose effects are not always easy to interpret. At the same time, every low-dimensional embedding necessarily distorts the original data to some degree (Chari and Pachter [2023]). A more transferable understanding of the algorithms is therefore needed so that practitioners can interpret embeddings cautiously, recognize spurious structure, and tune performance for new datasets. In practice, compact clusters may arise from initialization choices, learning-rate schedules, kernel parameters, or combinations of these design decisions. Yet the force dynamics that produce cluster formation have not been fully separated from these implementation details. Many influential parameters are also hidden inside abstract functional forms, which limits how broadly the resulting embeddings can be explained and compared.

In this paper, we separate the update forces into their constituent components and analyze the resulting functional shapes of attraction and repulsion. This decomposition shows that several behaviors often treated as method-specific—the need for learning-rate annealing, sensitivity to random initialization, and the emergence of clusters—are governed primarily by attraction. Repulsion, in contrast, mainly controls separation among clusters once attraction has organized local structure. Our specific contributions are:

1. We formulate attraction and repulsion shapes from the attractive and repulsive forces, establish general conditions for contraction and expansion of distance, and provide a transferable perspective on neighbor embedding algorithms (Section 4).
2. We show that UMAP’s attraction shape can induce both contraction and expansion of distance, and we use comparisons across algorithms to explain how attraction governs learning-rate annealing (Section 5).
3. We modify the attraction shape to improve embedding consistency under random initialization, indicating that a reproducible structure is encoded even when standard initialization is removed (Section 5.1).
4. We analyze repulsion shapes to clarify how cluster formation and inter-cluster distance are regulated across parameter choices (Section 5.2).
5. We compare attraction and repulsion shapes of UMAP, Parametric UMAP, and NEG-*t*-SNE, identify their shared structure and differences, and characterize their stability (Section 6).

The main text uses UMAP as the central case because it is a de facto standard in many fields, while the appendix extends the same lens to additional algorithms. The codes used in the research will be released upon acceptance.

2 Related Works

Modern iterative graph-based neighbor embedding algorithms trace back to stochastic neighbor embedding (SNE) (Hinton and Roweis 2002) and its *t*-distributed extension (*t*-SNE) (Maaten and Hinton, 2008). These methods use dense graphs in which every point has a pairwise relation with every other point, whether or not the points are close in the original space. Their graph weights are normalized so that the relations can be interpreted probabilistically. Other contemporaneous approaches, including locally linear embedding (Roweis and Saul 2000) and Laplacian Eigenmaps (LE) (Belkin and Niyogi 2002), instead used *k*-nearest neighbor (*k*-NN) graphs to encode pairwise interactions. As spectral methods, these algorithms rely on Eigenvalue decomposition rather than iterative attraction-repulsion updates. Tang et al. (Tang et al. 2016) later brought the *k*-NN graph into an iterative framework and removed normalization in the low-dimensional space. UMAP extended this line of work by removing normalization in both the high- and low-dimensional spaces and learning the embedding from pairwise interactions alone (McInnes et al. 2018). Its optimization explicitly combines an attractive force that preserves local neighborhoods with a repulsive force that separates dissimilar points. Methods such as PaCMAP (Wang et al. 2021) and NEG-*t*-SNE (Damrich et al. 2023) build on the same broad design space. For a recent survey of methods, see de Bodt et al. (2025).

Substantial work has sought principles that explain how these algorithms relate to one another. An early analysis of SNE showed that well-clustered data in the original space remain well clustered in the embedding space (Shaham and Steinerberger 2017). For *t*-SNE, Linderman and Steinerberger (2019) showed that the number of clusters in the embedding space lower-bounds the number of clusters in the original space, followed by further characterizations (Arora et al. 2018; Cai and Ma 2022; Linderman and Steinerberger 2022). Because *t*-SNE and UMAP arise from related frameworks while often producing different visualizations, a central question has been how to connect them (Böhm et al. 2022; Damrich et al. 2023; Draganov et al. 2023). Böhm et al. (2022) argued that Laplacian Eigenmap, ForceAtlas2 (Jacomy et al. 2014), UMAP, and *t*-SNE can be viewed as points on a common spectrum. Consistent with this view, LE and *t*-SNE are connected through the early exaggeration phase (Cai and Ma 2022), and UMAP and *t*-SNE can be related through contrastive estimation (Damrich et al. 2023). Around the same time, Hu et al. (2023) independently connected contrastive learning and SNE. Other recent explanations use probabilistic perspectives (Ravuri et al. 2023; Ravuri and Lawrence 2024), kernel techniques (Draganov and Dohn 2023), and information geometry (Kolpakov and Rocke 2024), underscoring that dimensionality reduction benefits from analyses that transfer across algorithms rather than treating each method in isolation.

3 Uniform Manifold Approximation and Projection

UMAP begins by constructing a high-dimensional graph for the dataset $X = \{x_i \in R^n | i = 1, \dots, N\}$ through pairwise affinities $p_{i,j} = f_h(d(x_i, x_j))$, typically with values in $[0, 1]$ (for details, see Appendix H).

The corresponding graph for the low-dimensional representation $Y = \{y_i \in \mathbb{R}^d | i = 1, \dots, N\}$ is defined by the differentiable affinity

$$q_{ij} = \frac{1}{1 + a(\|y_i - y_j\|_2^2)^b}, \quad (1)$$

where a and b set the density profile of the mapping and are chosen by fitting q_{ij} to

$$\Psi(d_{ij}) = \begin{cases} 1 & \text{if } \|y_i - y_j\|_2 < m_d, \\ \exp(-(d_{ij} - m_d)) & \text{otherwise} \end{cases}, \quad (2)$$

where m_d controls the desired spacing between nearest low-dimensional points. UMAP then minimizes the cross-entropy objective

$$\mathcal{L} = \sum_{i,j} (-p_{ij} \log(q_{ij}) - (1 - p_{ij}) \log(1 - q_{ij})). \quad (3)$$

The first term induces attraction and the second induces repulsion, a decomposition shared conceptually with several neighbor embedding methods. Rather than optimizing all pairs at every iteration, UMAP uses negative sampling (Mikolov et al., 2013; Tang et al., 2016): each edge with $p_{ij} > 0$ is treated as a positive edge, while several randomly sampled edges serve as negative edges. The attractive update is applied to the positive edge:

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(q_{ij}), \quad (4)$$

$$y_j^{t+1} = y_j^t + \lambda \nabla_{y_j^t} \log(q_{ij}), \quad (5)$$

and the repulsive update is applied to the negative edges:

$$y_i^{t+1} = y_i^t + \lambda \nabla_{y_i^t} \log(1 - q_{ij}), \quad (6)$$

where $\lambda(> 0)$ is the learning rate and t indexes the optimization step. For negative edges, only y_i is updated. For a detailed analysis of UMAP's loss, see Damrich and Hamprecht (2021).

4 Attraction and Repulsion Shapes

The updates in (4) and (6) can be expressed in a form that separates direction from strength. Specifically, each gradient $\nabla_{y_i^t} \log(q_{ij})$ or $\nabla_{y_i^t} \log(1 - q_{ij})$ decomposes into a distance-dependent scalar coefficient multiplying the displacement vector $(y_i - y_j)$. We call this scalar coefficient the attraction or repulsion shape. Although UMAP is the running example, the same decomposition applies to a broad class of embedding methods whose optimization is driven by attraction and repulsion.

Writing $\nabla_{y_i^t} \log(q_{ij}) = f_a(\zeta^t)(y_i^t - y_j^t)$, where $\zeta = \|y_i - y_j\|_2$, gives the following positive-edge updates:

$$y_i^{t+1} = y_i^t + \lambda f_a(\zeta^t)(y_i^t - y_j^t), \quad (7)$$

$$y_j^{t+1} = y_j^t - \lambda f_a(\zeta^t)(y_i^t - y_j^t). \quad (8)$$

Here, $f_a : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\leq 0}$ denotes the attraction shape, using the Euclidean identity $\nabla_{y_i^t} \log(q_{ij}) = -\nabla_{y_j^t} \log(q_{ij})$. The analogous negative-edge update can be written as

$$y_i^{t+1} = y_i^t + \lambda f_r(\zeta^t)(y_i^t - y_j^t), \quad (9)$$

$$y_j^{t+1} = y_j^t, \quad (10)$$

where $f_r : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is the repulsion shape.

Related decompositions have appeared before (McInnes et al., 2018; Agrawal et al., 2021; Draganov and Dohn, 2023), but they have been used for different purposes. The original UMAP paper (McInnes et al., 2018) used the form mainly as a computational device, whereas Draganov and Dohn (2023) used it to compare algorithms from a kernel perspective. In both cases, the derivative itself was the central object. Here, we instead treat the scalar decompositions as functions whose shapes can be analyzed, modified, and compared across methods. A similar perspective appears in Agrawal et al. (2021), but there the coefficient multiplies the unit vector $((y_i - y_j)/\|y_i - y_j\|_2)$ to emphasize force magnitude. The analysis below shows that the full shape, not only the magnitude, exposes behaviors that transfer across embedding algorithms.

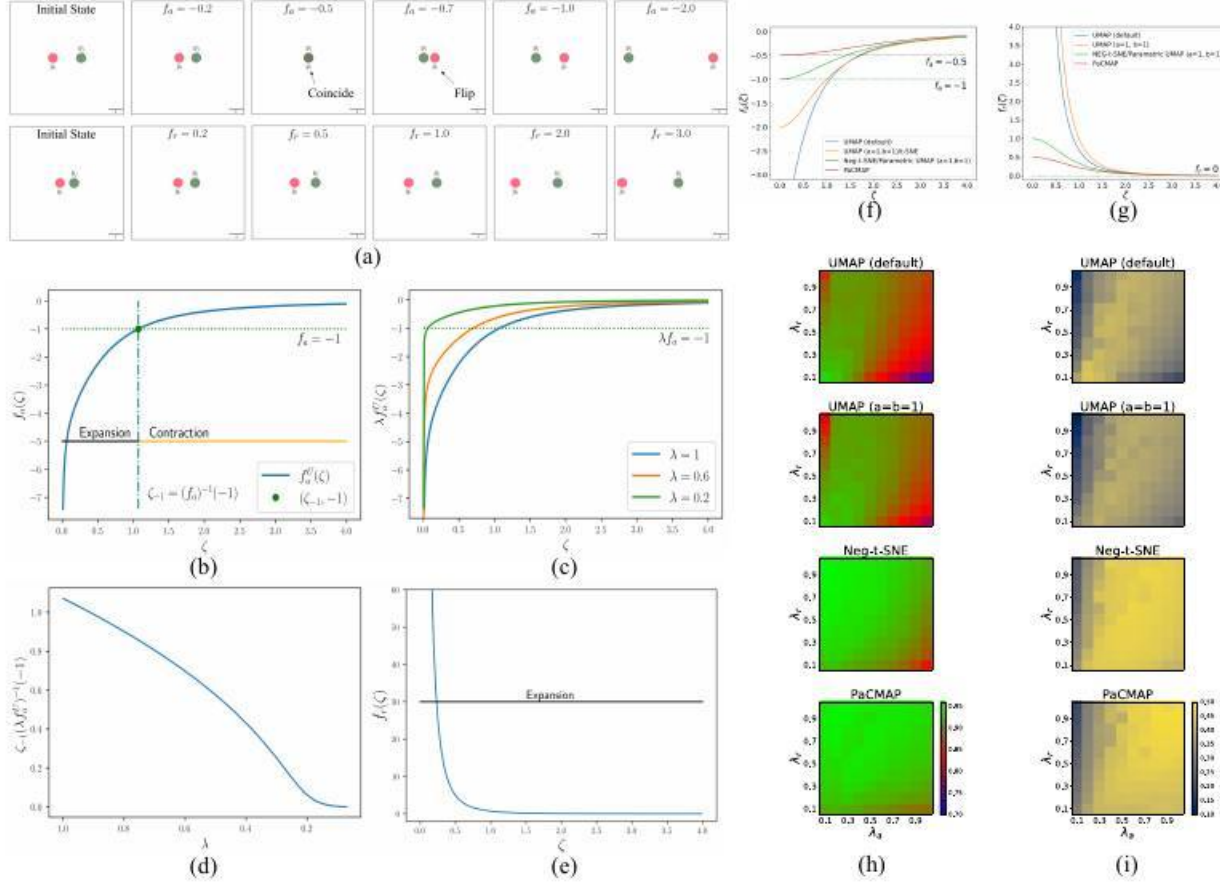


Figure 1: Attraction and repulsion shapes as a transferable description of UMAP-like updates. (a) Effects of representative values of f_a (top) and f_r (bottom) on a point pair. (b) UMAP attraction shape. (c) Effective attraction shape (λf_a) under different learning rates λ . (d) Minimum distance for contraction (ζ_{-1}) as λ decreases. (e) UMAP repulsion shape. (f) Attraction and (g) repulsion shapes for several embedding methods. (h) Trustworthiness and (i) Silhouette Score when constant attraction (λ_a) and repulsion (λ_r) learning rates are varied independently. Default UMAP parameters: $a = 1.58$ and $b = 0.89$.

4.1 Conditions for Attraction and Repulsion

We next derive general per-step conditions for attraction and repulsion from the update equations (7)-(10). The first proposition characterizes when a positive-edge update contracts the distance between y_i^t and y_j^t :

Proposition 4.1. *The update Eqs. (7) and (8) provide a contraction of distance ($\|y_i^{t+1} - y_j^{t+1}\| < \|y_i^t - y_j^t\|$) if $-1 < \lambda f_a < 0$.*

Here, λf_a is the effective attraction shape. In many comparisons, however, the unscaled function f_a already reveals the relevant embedding behavior. For a negative edge, the next proposition characterizes expansion of the distance between y_i^t and y_j^t :

Proposition 4.2. *The update Eqs. (9) and (10) provide an expansion of distance ($\|y_i^{t+1} - y_j^{t+1}\| > \|y_i^t - y_j^t\|$) if $f_r > 0$.*

Adding a symmetric term $-\lambda f_r(\zeta)(y_i^t - y_j^t)$ to Eq. (10) does not change this conclusion. These conditions should be read as per-iteration certificates for contraction or expansion, not as a restatement of learning-rate tuning. In particular, suitable shapes can make contraction and expansion behavior less dependent on learning-rate decay.

Figure 1(a) illustrates how different values of $f_a < 0$ and $f_r > 0$ affect a point pair. Repulsion is direct: any positive value of f_r increases the distance. Attraction is more nuanced because the same functional term can produce contraction, point flipping, or expansion. For $f_a \in (-1.0, 0)$, the distance decreases, with a sign flip at $f_a = -0.5$ where the

attractive update collapses the pair to coincident points. Any value below -1.0 increases the distance. Although these forces are local pairwise updates, their shape determines the global cluster structure that emerges across the embedding.

5 Analysis of UMAP in terms of Attraction and Repulsion Shape

Using the gradient decomposition and the distance form [1](#), the attraction and repulsion shapes are given by

$$f_a^U(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{1 + a\zeta^{2b}} \quad (11)$$

and

$$f_r^U(\zeta) = \frac{2b}{\zeta^{2b}(1 + a\zeta^{2b})}, \quad (12)$$

respectively. This section analyzes UMAP’s default shapes, governed primarily by a and b , and uses controlled perturbations of those shapes to identify behaviors that are not specific to one dataset. The arguments below apply for $b \leq 1$, where the attraction shape is strictly increasing and therefore invertible (for derivation, see [Appendix A](#)).

[Figure 1\(b\)](#) shows UMAP’s default attraction shape ($a = 1.58, b = 0.89$). The shape becomes unbounded as $\zeta \rightarrow 0$. By [Proposition 4.1](#) the transition from contraction to expansion occurs when λf_a^U crosses -1 at small ζ . Because the attraction shape is invertible, this transition defines $\zeta_{-1} = (\lambda f_a^U)^{-1}(-1)$, the minimum distance at which attractive updates still contract. Thus $\zeta > \zeta_{-1}$ yields contraction, whereas $0 < \zeta < \zeta_{-1}$ yields expansion, showing that nominal attraction does not universally pull neighboring points together.

When ζ_{-1} is large, neighboring points can oscillate between contraction and expansion, producing fuzzy cluster boundaries. Sharp boundaries therefore correspond to driving $\zeta_{-1} \rightarrow 0$, a goal that UMAP accomplishes through learning-rate annealing ([Figs. 1\(c,d\)](#)). In contrast, the repulsion shape in [Eq. 12](#) is always positive and satisfies [Proposition 4.2](#). Specifically, f_r^U approaches 0 as $\zeta \rightarrow \infty$ and approaches ∞ as $\zeta \rightarrow 0$ ([Fig. 1\(e\)](#)), making repulsion easier to interpret in this pairwise setting.

Comparing attraction shapes across algorithms shows that UMAP and t -SNE are the cases here in which $f_a < -1$ can arise (and consequently $\zeta_1 > 0$ at $\lambda = 1$, [Fig. 1\(f\)](#)). t -SNE mitigates this through update weights given by the corresponding p_{ij} values, whereas UMAP relies on learning-rate annealing. Neg- t -SNE and PaCMAP keep f_a naturally within $[-1, 0]$, satisfying [Proposition 4.1](#) for $\lambda \in [0, 1]$ with $\zeta_{-1} = 0$. PaCMAP’s weighted f_a is always below 0.5, which further prevents flips during attraction. Repulsion shapes provide a complementary comparison: among the methods shown, UMAP is the one with very large repulsive values at small distances ([Fig. 1\(g\)](#)); the shape is unbounded, although implementations often clip these values during optimization. For additional details of these and related methods, see [Appendix G](#).

We evaluate these attraction/repulsion choices on MNIST by assigning separate constant learning rates to attraction (λ_a) and repulsion (λ_r) and varying them independently ([Figs. 1\(h,i\)](#)). For UMAP, smaller λ_a values (< 0.5 and consequently $\zeta^{-1} \approx 0$) give better embeddings by trustworthiness (T) ([Venna and Kaski, 2001](#)), a measure of structure preservation ([Fig. 1\(h\)](#)), and better label separation by silhouette score (SIL) ([Rousseeuw, 1987](#)) ([Fig. 1\(i\)](#)). By contrast, the full tested range $\lambda_r \in [0, 1]$ remains usable. For NEG- t -SNE and PaCMAP, where $\zeta_{-1} = 0$, the full parameter range is effective in these experiments. These results support the broader interpretation that UMAP’s stable embedding and clustering behavior depends on making ζ_{-1} close to 0, which the standard algorithm achieves through learning-rate annealing; newer algorithms can instead build this property into the default attraction shape. For additional discussions, results, and illustrations of these methods on different datasets, see [Section 6](#) and [Appendices B, C](#) and [I](#).

5.1 Improving UMAP’s Consistency under Random Initialization

UMAP’s consistency depends strongly on initialization ([Kobak and Linderman, 2021](#), [Wang et al., 2021](#)), a concern that applies whenever embeddings are used as reproducible summaries rather than one-off visualizations. In standard practice, PCA of the data or spectral decomposition of the high-dimensional graph initializes the embedding, which helps produce consistent mappings despite stochastic optimization. With random initialization, clusters may fail to form or may appear in a different orientation on each run. The shape perspective explains this behavior: if two high-dimensional nearest neighbors begin far apart in the embedding, attraction can be too weak to bring them together. This near-sightedness ([Wang et al., 2021](#)) is visible directly in the attraction shape, where $|f_a^U|$ diminishes toward zero as distance increases (i.e., $\lim_{\zeta \rightarrow \infty} |f_a^U(\zeta)| = 0$).

A natural way to counter this limitation is to make the embedding more “far-sighted” by increasing attraction at large distances, allowing distant high-dimensional neighbors to approach each other in the low-dimensional space. To test

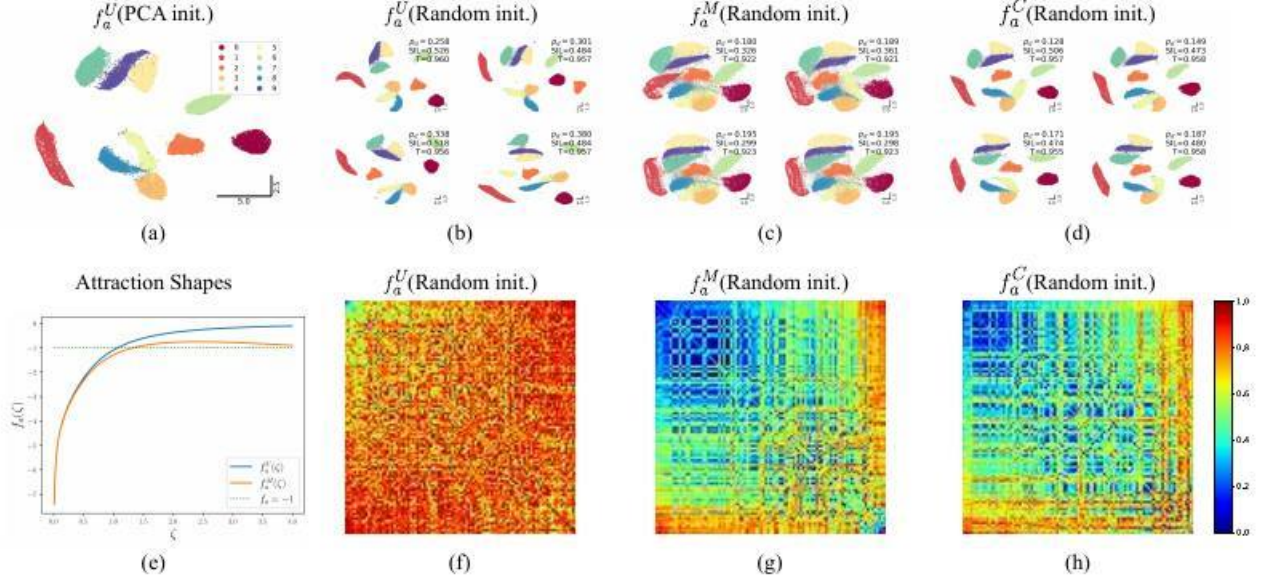


Figure 2: Random-initialization behavior under different attraction shapes on MNIST. (a) PCA-initialized reference mapping. (b-d) Four random-initialized mappings with the lowest Procrustes distance (p_d) from (a) for (b) UMAP, (c) modified attraction, and (d) composite attraction. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices from 100 runs for (f) UMAP (0.78 ± 0.13), (g) modified attraction (0.49 ± 0.21), and (h) composite attraction (0.50 ± 0.20). Diagonal (i, i) entries are sorted by p_d from (a), while off-diagonal (i, j) entries give p_d between the i^{th} and j^{th} mappings. These matrices and (mean $p_d \pm \text{std}$) values show that default UMAP embeddings are not mutually self-similar under random initialization, whereas the modified and composite attraction shapes recover more scale-invariant structure.

this hypothesis within the same UMAP framework, we modify the attraction shape as follows:

$$f_a^M = f_a^U - \beta \zeta, \quad (13)$$

where β controls the strength of the added term (we used $\beta = 0.2$, Fig. 2(e)). At the loss level, this shape modification corresponds to adding a regularizer to the attractive term (i.e., $\mathcal{L}^M = \mathcal{L} + \sum_{i,j} \beta/3 \|y_i - y_j\|_2^3$). Beyond pulling distant neighbor pairs closer, the modification promotes point intermixing that helps convergence from random initialization, analogous in role to early exaggeration in t -SNE. Equation (13) uses the simplest linear correction; other distance-dependent functions, such as $\log \zeta$ or ζ^p ($p \in \mathbb{R}_{\geq 0}$), may serve the same broader purpose.

We also consider a composite attraction shape:

$$f_a^C = \begin{cases} f_a^M, & \text{epoch} \leq 100 \\ f_a^U, & \text{otherwise} \end{cases}. \quad (14)$$

The composite shape uses the modified attraction only early in optimization and then returns to the original UMAP shape, aiming to retain the benefits of far-sighted attraction while limiting later distortion. We next evaluate how the modified and composite shapes affect dimensionality reduction from random initialization.

We first construct a PCA-initialized MNIST embedding as the reference in Fig. 2(a). For each attraction shape, we then generate Gaussian-random-initialized embeddings and repeat the experiment 100 times. We quantify run-to-run consistency with Procrustes analysis (Gower [1975]), which aligns point clouds under scaling, translation, rotation, and reflection (for details, see Appendix D.1). In this experiment, each random-initialized embedding is aligned to the PCA-initialized reference and compared using Procrustes distance (p_d). Figure 2(b) shows the four default-UMAP embeddings with the lowest p_d : the cluster shapes are broadly consistent, but their placements vary. The modified and composite attraction shapes (Figs. 2(c) and (d)) produce more consistent cluster placements under the same random-initialization protocol.

We further summarize placements with a Procrustes matrix: diagonal entries are sorted by p_d from the PCA-initialized mapping, and off-diagonal entries are p_d values between pairs of random-initialized mappings (for details, see Appendix D.1). This representation is analogous to a similarity matrix (Foote [1999]). Default UMAP attraction yields embeddings that are not mutually similar (Fig. 2(f)), whereas the modified (Fig. 2(g)) and composite (Fig. 2(h))

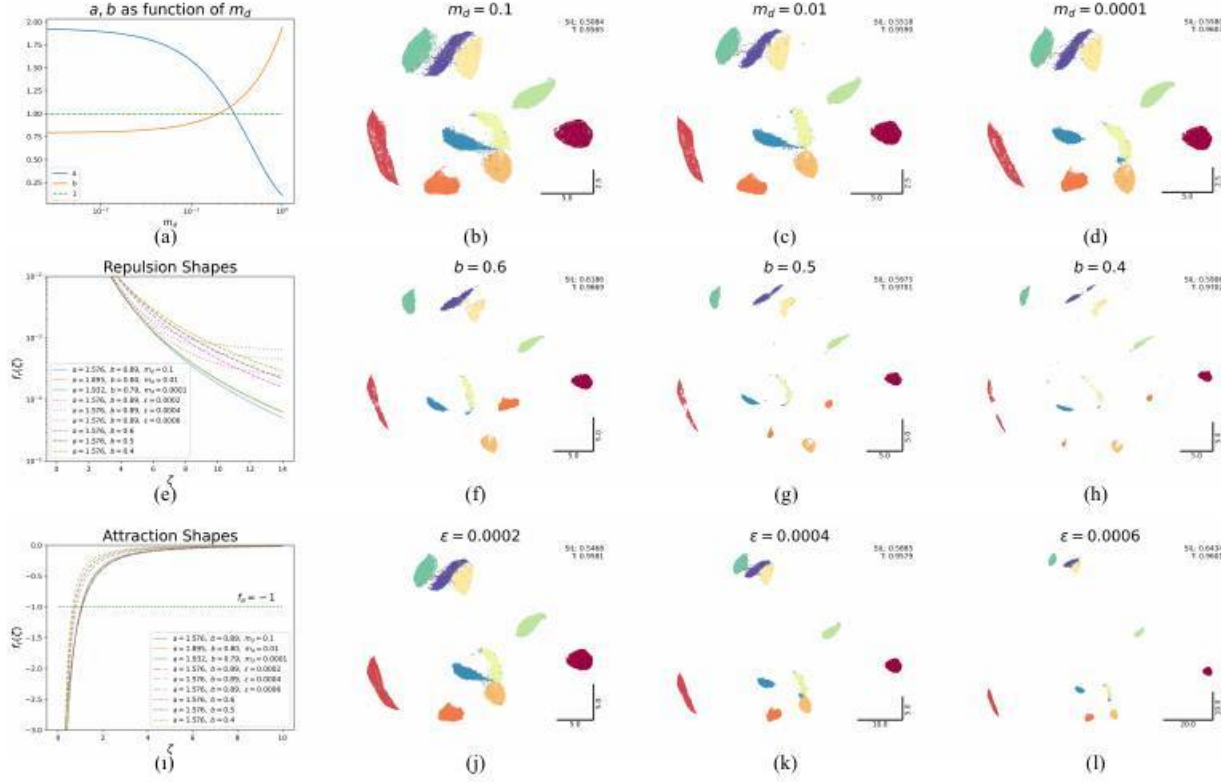


Figure 3: Controlling inter-cluster distances on MNIST through parameter and shape changes. (a) Computing a , b through the low-dimensional distance m_d restricts the explored shape range. (b-d) UMAP embeddings with m_d set to 0.1, 0.01, and 0.0001 show limited gains in cluster compactness. (e) Repulsion shapes under different parameters. (f-h) Increasing repulsion by explicitly varying b yields more compact clusters and separates structures that were not distinct before. (i) Attraction shapes under parameter variation. (j-l) Adding a small positive value (ϵ) to the repulsion shape increases inter-cluster distance while keeping attraction fixed.

shapes produce strong mutual similarity. This suggests that UMAP encodes a reproducible structure that is not fully recovered from random initialization when long-distance attraction is too weak.

5.2 Cluster Formation and Compactness

In standard UMAP usage, the minimum distance parameter m_d is the most visible control over cluster formation (through Eq. 2). However, changing m_d explores only a restricted subset of the possible a and b values (Fig. 3(a)). Consequently, reducing m_d often yields embeddings with little additional benefit (Figs. 3(b-d)). The shape analysis explains why: varying m_d has limited effect on the repulsion shape (Fig. 3(e)). A broader intervention is to vary a and b directly. Decreasing a strengthens repulsion, but it weakens attraction even faster and therefore worsens near-sightedness. Decreasing b provides more useful control over both shapes (Figs. 3(e,i)). Figures 3(f-h) show that setting b to 0.6, 0.5, and 0.4 progressively increases inter-cluster distance and separates clusters that were previously merged. This separation is consistent with the increasingly heavy-tailed kernel induced by smaller b values; heavy-tailed kernels tend to produce smaller and more distinct clusters (Van der Maaten and Hinton, 2008; Kobak et al., 2019), and Kobak et al. (2019) discusses this tail modulation in UMAP. Although this parameter change separates all ten MNIST labels into distinct clusters, it simultaneously alters attraction and repulsion, so the respective contributions of the two forces remain entangled.

To isolate the effect of repulsion, we keep attraction fixed and modify only the repulsion shape by adding a small positive value (ϵ):

$$f_r^M = f_r^U + \epsilon, \quad (15)$$

while holding a and b fixed. This modification is equivalent to adding a regularizer to the repulsive term of the loss (i.e., $\mathcal{L}^M = \mathcal{L} - \sum_{i,j} \epsilon/2 \|y_i - y_j\|_2^2$) and gives stronger repulsion than the parameter settings above (Fig. 3(e)).

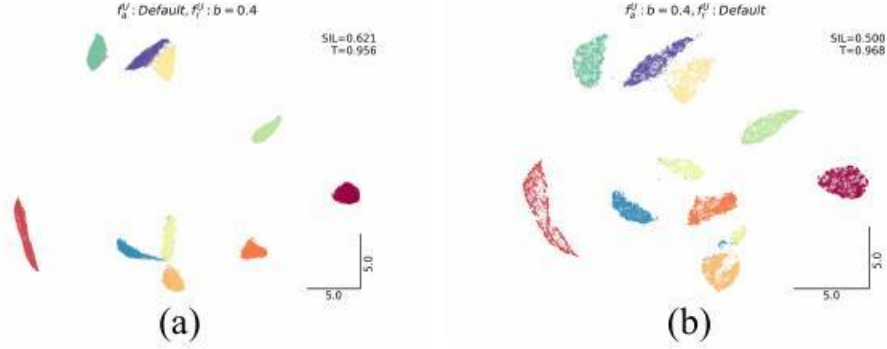


Figure 4: MNIST embeddings that isolate attraction and repulsion changes. (a) Default attraction with the repulsion shape induced by $b = 0.4$ preserves the default UMAP clusters while increasing compactness. (b) Default repulsion with the attraction shape induced by $b = 0.4$ reveals additional within-cluster structure and forms new clusters.

Figs. 3(j-l) show that larger ε increases inter-cluster distances. The resulting cluster patterns remain similar to those obtained by varying m_d , producing loose separation of all labels as ε increases. Because ε changes repulsion without changing attraction, and because varying m_d traces similar attraction shapes (Fig. 3(i)), these results suggest that cluster formation is governed primarily by attraction, while repulsion adjusts spacing.

We further disentangle cluster formation by changing either attraction or repulsion while leaving the other at its default, using the shapes induced by $b = 0.4$ (Fig. 4). With default attraction and increased repulsion, the embedding preserves the default UMAP cluster structure but makes the clusters smaller (Fig. 4(a)). With increased attraction and default repulsion, the embedding reveals additional structures within clusters (Fig. 4(b)). Some previously merged clusters split into smaller groups. These controlled comparisons support a general division of labor: attraction drives the formation of clusters and substructure, whereas repulsion controls compactness either by shrinking clusters or increasing inter-cluster distance. Appendix G.3 shows that LocalMAP embeddings exploit a related attraction-repulsion interplay to separate clusters.

6 Comparison to NEG- t -SNE

The setting $a = 1$ and $b = 1$ appears in several dimensionality reduction algorithms because it often yields stable, bounded gradients and therefore simpler optimization. Damrich et al. (2023) examined this setting for UMAP and proposed Neg- t -SNE, which combines stable gradients with more compact clustering even when the learning rate is fixed at 1. Parametric UMAP (Sainburg et al., 2021) began from the UMAP loss, but later it adopted a numerically stable modified cross-entropy loss (Shi et al., 2023) with a logsigmoid kernel. Analyzing these methods through shapes shows that they reach the same formulation from different starting points, yielding the following observation:

Proposition 6.1. *Neg- t -SNE is Parametric UMAP with $a = 1$ and $b = 1$.*

For UMAP with $a = 1$ and $b = 1$, the attraction and repulsion shapes become $f_a^U = -2/(1 + \zeta^2)$ and $f_r^U = 2/(\zeta^2(1 + \zeta^2))$, respectively. The attraction shape is bounded in $[-2, 0]$, with $f_a^U(0) = -2$ (Fig. 1(f)), while the repulsion shape remains unbounded as $\zeta \rightarrow 0$ (Fig. 1(g)). Because $f_a^U < -1$ near $\zeta = 0$, Proposition 4.1 and the analysis in Section 5 imply that even this simplified UMAP case still needs learning-rate annealing.

From the contrastive embedding (CE) perspective, Damrich et al. (2023) showed that UMAP’s negative-sampling loss has effective kernel $1/\zeta^2$. They introduced NEG- t -SNE by replacing this kernel with $1/(1 + \zeta^2)$. In UMAP notation, the resulting low-dimensional affinity is $q_{ij}^N = 1/(2 + \zeta^2)$, giving attraction and repulsion shapes $f_a^N = -2/(2 + \zeta^2)$, and $f_r^N = 2/((1 + \zeta^2)(2 + \zeta^2))$, respectively (Figs. 1(f,g)). The attraction shape lies in $[-1, 0]$ and therefore satisfies Proposition 4.1. Any $\lambda \in [0, 1]$ produces contraction without oscillating between expansion and contraction. Thus, NEG- t -SNE is less sensitive to learning-rate annealing, and its clusters remain less fuzzy even at constant $\lambda = 1.0$. Its repulsion shape is also bounded within $[0, 1]$ and does not diverge as $\zeta \rightarrow 0$, reducing the tendency for randomly sampled points to leave clusters. Whereas Damrich et al. (2023) explained the disparity through bounded NEG- t -SNE repulsion and unbounded UMAP repulsion, our shape analysis in Fig. 1 shows that attraction also contributes directly to stability and compactness. For additional discussion on NEG- t -SNE with illustration, see Appendix F. Together with

Fig. 4 this suggests a more general design principle: attraction and repulsion shapes from different methods can be recombined, for example by pairing UMAP attraction with NEG- t -SNE repulsion. Appendix J gives examples using UMAP, NEG- t -SNE, and PaCMAP.

7 Discussion and Conclusion

This work studies how attractive and repulsive forces shape cluster formation in neighbor embedding algorithms. Although attraction is usually described as bringing similar points together and repulsion as pushing dissimilar points apart, the mechanisms by which these forces create visual structure are more varied. By analyzing UMAP and NEG- t -SNE in the main text and additional algorithms in Appendix G we show that attraction-repulsion shapes provide a broadly applicable framework for interpreting dimensionality reduction behavior within the settings studied here.

We treat attraction and repulsion coefficients as functional mappings and study their shapes directly. The resulting characterizations (Propositions 4.1 and 4.2) reveal a counterintuitive behavior: UMAP’s attractive update can expand distances at short separations (Fig. 1 (b)), so an attraction term can push neighboring points apart in that regime. UMAP’s learning-rate schedule mitigates this behavior by moving the effective attraction shape toward the contraction regime.

We also examine initialization as a general source of variability in low-dimensional embeddings. Similar points that start far apart experience weak attraction and may never contract sufficiently. The shape formalism makes it possible to strengthen long-distance attraction by adding terms to the attraction shape. Under random initialization, this modification produces more consistent outputs, supporting the view that UMAP encodes a reproducible structure that standard initialization and scheduling help recover.

The repulsion analysis shows that stronger repulsion increases inter-cluster distance. Adding a small positive value to the repulsion shape provides a direct way to obtain stronger separation than is accessible through the usual minimum-distance parameter m_d alone.

The comparison between NEG- t -SNE and UMAP further illustrates how shape constraints transfer across model classes. For $a = 1$ and $b = 1$, NEG- t -SNE differs primarily by providing bounded attraction ($\in [-1, 0]$) and repulsion ($\in [0, 1]$) shapes. Both propositions then hold across the tested fixed learning rates $\in [0, 1]$, reducing sensitivity to annealing. Allowing a and b to vary in NEG- t -SNE reintroduces the numerical challenges seen in UMAP. We also show that NEG- t -SNE and Parametric UMAP are equivalent under the stated formulation.

These insights offer practical tools for tuning dimensionality reduction algorithms and conceptual tools for interpreting their outputs across datasets and embedding objectives. Because dimensionality reduction and contrastive learning are closely connected (Damrich and Hamprecht 2021, Hu et al. 2023), the same shape-based perspective may also inform representation learning. Overall, the paper frames embedding forces as analyzable design objects, making visualizations and their interpretations more principled, consistent, and reliable.

Software and Data

All datasets used in the experiments are publicly available, supporting reuse of the same image, transcriptomic, and document-embedding settings studied here. The MNIST dataset is available at yann.lecun.com. Fashion-MNIST is available at github.com. Single-cell transcriptomes data is available at github.com.

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Appendix

Appendix [A](#) provides the derivations for UMAP attraction and repulsion shapes. Appendix [B](#) revisits constant learning rates through the same shape lens, and Appendix [C](#) applies the attraction-repulsion interpretation to large language model embeddings. Appendix [D.1](#) defines the Procrustes distance and matrix used to quantify consistency, while Appendix [E](#) repeats the random-initialization experiment on additional datasets. Appendix [F](#) extends the comparison between NEG-*t*-SNE and UMAP, and Appendix [G](#) applies the analysis to other dimensionality reduction algorithms. Appendix [H](#) summarizes high-dimensional graph construction, Appendix [I](#) provides detailed λ_a and λ_r sweeps from Fig. [I](#) for UMAP, NEG-*t*-SNE, and PaCMAP on multiple datasets, Appendix [J](#) tests mixed attraction and repulsion shapes from different algorithms, and Appendix [K](#) gives implementation details.

A Proofs and Derivations

A.1 Proof of Proposition [4.1](#)

Proof. We subtract Eq. [\(7\)](#) from Eq. [\(8\)](#) and take a norm:

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + 2\lambda f_a| \|y_i^t - y_j^t\|. \quad (16)$$

This distance contracts as long as $|1 + 2\lambda f_a| < 1$, i.e., provided

$$-1 < \lambda f_a < 0, \quad (17)$$

□

A.2 Proof of Proposition [4.2](#)

Proof. We subtract Eq. [\(9\)](#) from Eq. [\(10\)](#) and take a norm:

$$\|y_i^{t+1} - y_j^{t+1}\| = |1 + \lambda f_r| \|y_i^t - y_j^t\|. \quad (18)$$

This distance increases when $|1 + \lambda f_r| > 1$, i.e.,

$$\lambda f_r < -2 \text{ or } f_r > 0. \quad (19)$$

From the definition of f_r , the latter suffices. □

A.3 Attraction Shape:

We use a general form of the low-dimensional affinity function, i.e., $q_{ij} = (\gamma + a\|y_i - y_j\|_2^{2b})^{-1}$, to derive the attraction shape. It reduces to UMAP for $\gamma = 1$ and to NEG-*t*-SNE for $\gamma = 2$. The attractive force is given by

$$\begin{aligned} \nabla_{y_i} \log q_{ij} &= -\nabla_{y_i} \log (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} (\gamma + a(\|y_i - y_j\|_2^2)^b) \\ &= -\frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} ab(\|y_i - y_j\|_2^2)^{b-1} \nabla_{y_i} \|y_i - y_j\|_2^2 \\ &= -\frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} (y_i - y_j). \end{aligned} \quad (20)$$

Defining $\zeta = \|y_i - y_j\|_2$, the first term gives the attraction shape as:

$$f_a(\zeta) = -\frac{2ab\zeta^{2(b-1)}}{\gamma + a\zeta^{2b}} \quad (21)$$

A.4 Condition for strictly increasing f_a^U :

Its behavior with distance can be characterized by computing the derivative of f_a^U :

$$\frac{df_a^U(\zeta)}{d\zeta} = -\frac{2ab\zeta^{2b-3}}{\gamma + a\zeta^{2b}} \left((b-1) - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}} \right). \quad (22)$$

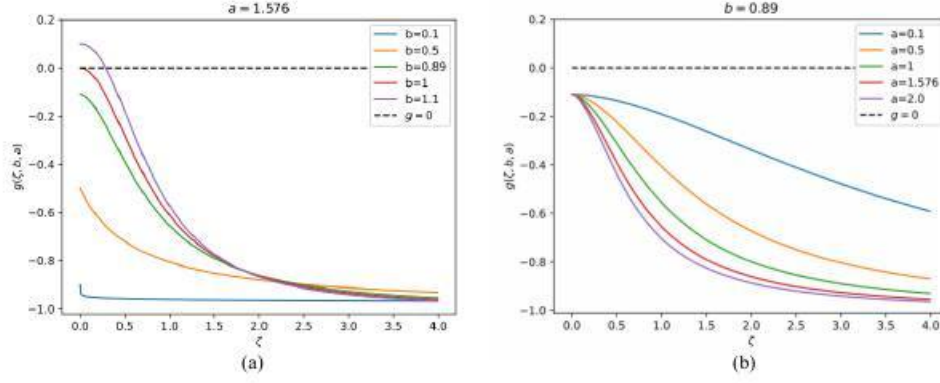


Figure 5: Values of $g(\zeta, b, a)$ for (a) a fixed at 1.576 and (b) b fixed at 0.89.

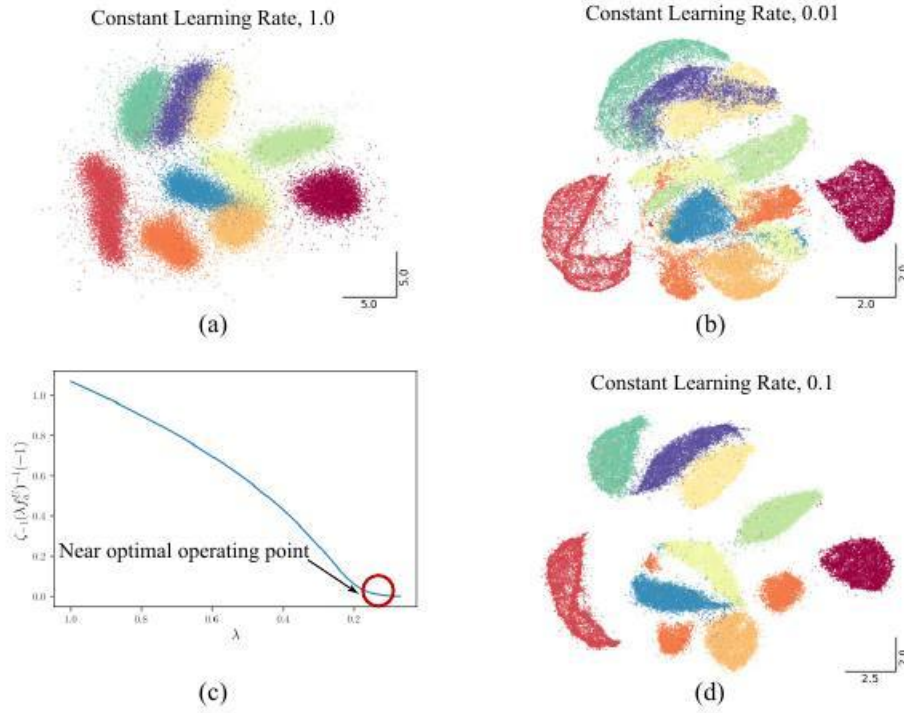


Figure 6: Effect of constant learning rate in embeddings. (a) When the learning rate is too high ($\lambda = 1.0$), the embeddings are diffuse (because of the high value of ζ_{-1}). (b) When the learning rate is too low ($\lambda = 0.01$), clusters don't form (the strengths of attraction and repulsion are too low). (c) ζ_{-1} decreases nonlinearly as the learning rate decreases. The goal of the algorithm is to reduce ζ_{-1} while keeping effective levels of attraction and repulsion. (d) Distinct and compact clusters form at a constant, near-optimal learning rate $\lambda = 0.1$.

This leads to a strictly increasing condition ($\frac{df^U}{d\zeta} > 0$),

$$g(\zeta, b, a) < 0, \quad (23)$$

where $g(\zeta, b, a) = b - 1 - \frac{ab\zeta^{2b}}{\gamma + a\zeta^{2b}}$. This inequality is valid as long as $0 < b \leq 1$ (using the derivative and asymptotes of g). Figure 5 shows values of g for different b and a . As b increases above 1, the inequality (23) no longer holds.

A.5 Repulsion Shape:

The repulsive force, using the general form of the low-dimensional affinity, is given by

$$\begin{aligned}
 \nabla_{y_i} \log(1 - q_{ij}) &= \nabla_{y_i} \log \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\
 &= \frac{\gamma + a(\|y_i - y_j\|_2^2)^b}{(\gamma - 1) + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \left[1 - \frac{1}{\gamma + a(\|y_i - y_j\|_2^2)^b} \right] \\
 &= \frac{1}{\gamma - 1 + a(\|y_i - y_j\|_2^2)^b} \frac{ab(\|y_i - y_j\|_2^2)^{b-1}}{\gamma + a(\|y_i - y_j\|_2^2)^b} \nabla_{y_i} \|y_i - y_j\|_2^2 \\
 &= \frac{2ab(\|y_i - y_j\|_2^2)^{b-1}}{(\gamma - 1 + a(\|y_i - y_j\|_2^2)^b)(\gamma + a(\|y_i - y_j\|_2^2)^b)} (y_i - y_j).
 \end{aligned} \tag{24}$$

The scalar coefficient in this expression gives the repulsion shape:

$$f_r(\zeta = \|y_i - y_j\|_2) = \frac{2ab\zeta^{2(b-1)}}{(\gamma - 1 + a\zeta^{2b})(\gamma + a\zeta^{2b})}. \tag{25}$$

For the parameter ranges considered here, $f_r > 0$, so the term behaves as a repulsive shape.

A.6 Loss functions due to modified attraction and repulsion shape:

The cost function of the attractive term with the modification in Eq. (13) is given by

$$-\int (f_a^U(\|y_i - y_j\|_2) - \beta\|y_i - y_j\|_2)(y_i - y_j) dy_i = -\log(q_{ij}) + \frac{\beta}{3}\|y_i - y_j\|_2^3, \tag{26}$$

whereas the repulsive term due to Eq. (15) is given by

$$-\int (f_r^U(\|y_i - y_j\|_2) + \epsilon)(y_i - y_j) dy_i = -\log(1 - q_{ij}) - \frac{\epsilon}{2}\|y_i - y_j\|_2^2. \tag{27}$$

In both cases, the added component is a norm-based regularizer. Expressing the modification at the shape level makes the effect more interpretable: the attractive term increases far-sightedness, whereas the repulsive term adds a constant repulsive coefficient.

B More on UMAP’s Learning Rate

Sections 5 and 6 explain UMAP’s reliance on learning-rate annealing through the shape parameter ζ_{-1} (Fig 6). In Section 5, ζ_{-1} is defined as the minimum distance for contraction, and reducing it through annealing produces compact clusters. Section 6 compares UMAP with $a = 1.0$ and $b = 1.0$ to Neg- t -SNE and shows that Neg- t -SNE tolerates a constant learning rate of 1.0 better because its attraction shape lies in $[-1, 0]$, whereas UMAP’s lies in $[-2, 0]$. The same reasoning shows that UMAP can tolerate a constant learning rate of 0.5 when the effective attraction shape is confined to $[-1, 0]$ (Figs. 11(f,h)).

Even so, annealing remains beneficial for a wide range of a and b values. The reason is that the algorithm ultimately benefits from driving ζ_{-1} toward zero, which occurs as the learning rate approaches zero. If this does not happen, the embedding remains diffuse (Fig 6(a)). If the learning rate starts too low, however, attraction and repulsion are both too weak to form clear clusters (Fig 6(b)). The ζ_{-1} versus λ curve (Fig 6(c)) shows that $\lambda = 0.1$ is a useful constant-learning-rate compromise, with low ζ_{-1} and still appreciable attraction and repulsion. Using $\lambda = 0.1$ yields compact clusters with clear boundaries in this experiment (Fig 6(d)).

C Large Language Model Embedding

Stronger attraction and repulsion can also improve embeddings outside image benchmarks. As an additional application scenario, we randomly selected 300,000 PubMedBERT feature vectors (Gu et al., 2021) with known labels from the PubMed dataset (González-Márquez et al., 2024). Default UMAP produces a crowded map in which label colors overlap (Fig. 7(a)). Without labels, this embedding is difficult to interpret because explicit clusters are weak. Reducing b to 0.6 (as discussed in Section 5.2) increases repulsion while also supporting cluster formation through improved attraction. The resulting map is more explorable even without explicit labels (Fig. 7(b)). We quantify the improvement with k-NN accuracy ($k = 10$). On a hold-out set of 10,000 points, reducing b to 0.6 increases k-NN accuracy from 49.2% to 55.48%.

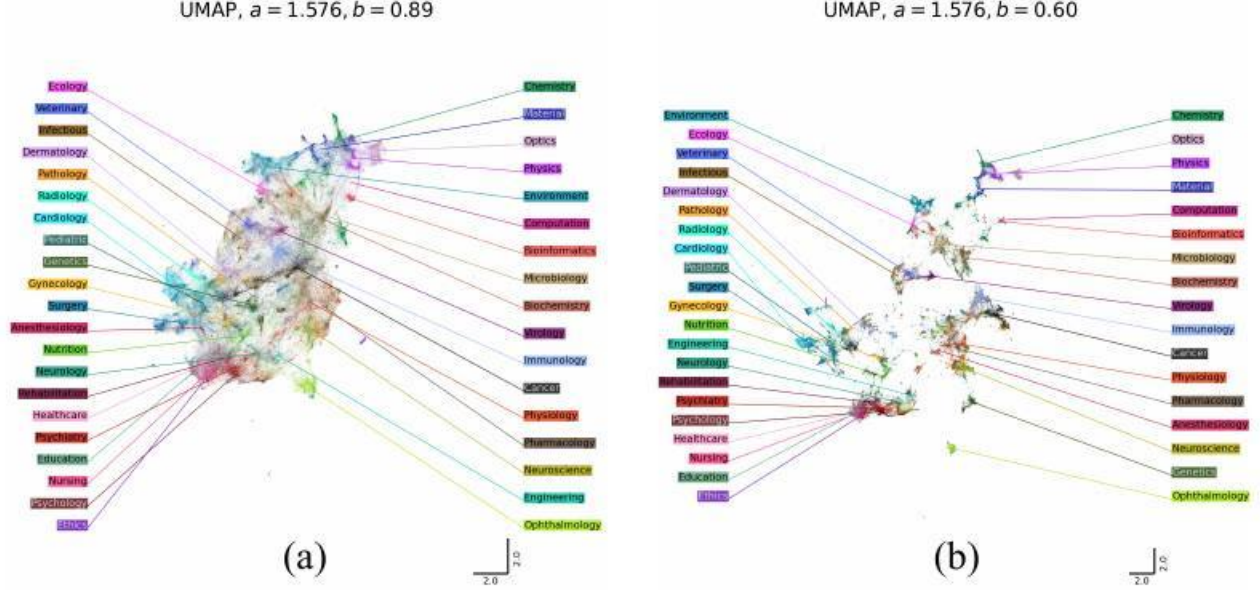


Figure 7: Visualization of 300K PubMed abstracts embedded from PubMedBERT features. (a) Default UMAP ($a = 1.576, b = 0.89$) produces overlapping label regions. (b) UMAP with stronger attraction-repulsion effects through $a = 1.576, b = 0.60$ produces clearer cluster formation and less label overlap. Labels are shown in both panels for interpretation.

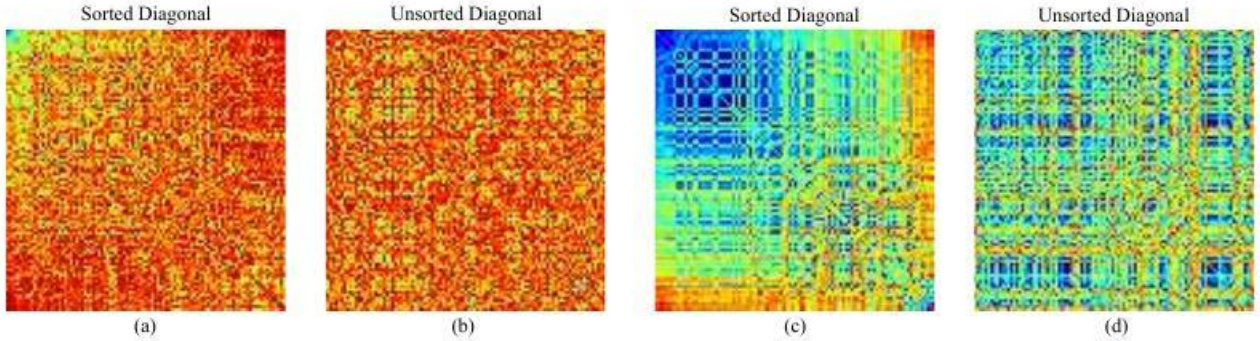


Figure 8: Effect of diagonal sorting on Procrustes-matrix visualization. (a) Procrustes matrix from Fig. 2f for default UMAP attraction, with the diagonal sorted by p_d from a PCA-initialized reference embedding. (b) The same data as in (a), but with the diagonal unsorted or randomly sorted. (c) Procrustes matrix from Fig. 2g for modified attraction, again sorted by p_d from the PCA-initialized reference. (d) The same data as in (c), but with the diagonal unsorted.

D Metrics

D.1 Procrustes Distance and Procrustes Matrix

The Procrustes distance (Gower 1975) measures similarity between two point clouds $\{x\}$ and $\{y\}$ after allowing linear transformations such as translation, scaling, and rotation. Operationally, one point cloud is held fixed while the other is transformed to maximize alignment. Let $\{y'\}$ denote the transformed version of $\{y\}$ that achieves this alignment. The Procrustes distance is then

$$p_d(\{x\}, \{y\}) = \sqrt{\sum_k (x_k - y'_k)^2}. \quad (28)$$

This linear comparison has been useful across several embedding and alignment settings (McInnes et al. 2018; Islam and Fleischer 2022; Kotlov et al. 2024).

We use Procrustes distance to quantify embedding consistency under random initialization. Let $\{x\}_p$ be a reference embedding, obtained with PCA initialization in our experiments, and let $X_r = \{\{x\}_i | i = 1, 2, 3, \dots, N\}$ be the set of N embeddings obtained from random initialization. We summarize mutual similarity by taking the mean and standard deviation of the strictly lower-triangular entries of P (reported as mean $\pm$ std in Figs. 2, 9 and 10), with

$$\text{mean} = \frac{2}{N(N-1)} \sum_{i,j(i>j)} P_{i,j}, \text{ and } \text{std} = \sqrt{\sum_{i,j(i>j)} \frac{2(P_{i,j} - \text{mean})^2}{N(N-1)}} \quad (29)$$

The indexes of X_r can be sorted such that $p_d(\{x\}_i, \{x\}_p) \leq p_d(\{x\}_{i+1}, \{x\}_p)$, so that the diagonal values of the Procrustes matrix are given by

$$P_{i,i} = p_d(\{x\}_i, \{x\}_p) \quad (30)$$

and the off-diagonal values are given by

$$P_{i,j} = p_d(\{x\}_i, \{x\}_j). \quad (31)$$

Sorting the diagonal does not change the numerical values, but it makes self-similar embedding families easier to see. In Fig. 8(a), for example, the sorted diagonal reveals a block of similar embeddings in the upper-left region of the matrix. With the modified attraction shape, this mutually similar region grows (the larger blue region in Fig. 8(c)), suggesting that the optimization has a more accessible metastable configuration. When the diagonal is unsorted, the same structure is visually obscured and the similarity pattern is harder to interpret (Fig. 8(b,d)).

D.2 Trustworthiness

The trustworthiness metric (Venna and Kaski 2001) quantifies how well local neighborhoods are preserved after dimensionality reduction:

$$T = 1 - \frac{2}{nk(2n - 3k - 1)} \sum_{i=1}^n \sum_{y_j \in \text{KNN}(y_i, k)} \max(0, r(i, j) - k) \quad (32)$$

where $\text{KNN}(y_i, k)$ is the k -NN graph in the embedding space and $r(i, j)$ is the rank of x_j in the high-dimensional k -NN graph. In practice, k is often set to 5, so the metric evaluates preservation of each point's five nearest neighbors. For computational efficiency, all reported trustworthiness values use 10,000 randomly sampled indices, with the same indices used when comparing embeddings.

D.3 Silhouette Score

Although the silhouette score (Rousseeuw 1987) was designed for evaluating clustering algorithms, we use it here as a label-separation measure in the embedding. This is appropriate for our benchmarks because the embedding methods naturally form clusters, and the ground-truth labels provide an external way to assess their separation. For a point y_i in a point cloud $\{y\}$, let a_i be the mean distance from y_i to other points with the same label, and let b_i be the smallest mean distance from y_i to points with any other label. The pointwise silhouette is

$$s_i = \frac{b_i - a_i}{\max\{a_i, b_i\}} \quad (33)$$

This value lies in $[-1, 1]$. Values near 1 indicate that y_i lies well within its label cluster, values near 0 indicate a boundary point, and values near -1 indicate poor label separation. The overall silhouette score is

$$\text{SIL} = \frac{1}{N} \sum s_i. \quad (34)$$

We compute the silhouette score on the full embedding, without random sampling, using Euclidean distances.

E Effect of Random Initialization in Additional Datasets

Section 5.1 presents the random-initialization experiment on MNIST. To test whether the same pattern appears beyond that benchmark, we repeat the experiment on Fashion-MNIST (FMNIST) (Xiao et al. 2017) and single-cell transcriptomes (Macosko et al. 2015) (Fig. 9 and 10 respectively). The conclusion is consistent across these datasets: modified and composite attraction shapes that strengthen long-distance attraction substantially improve reconstruction consistency.

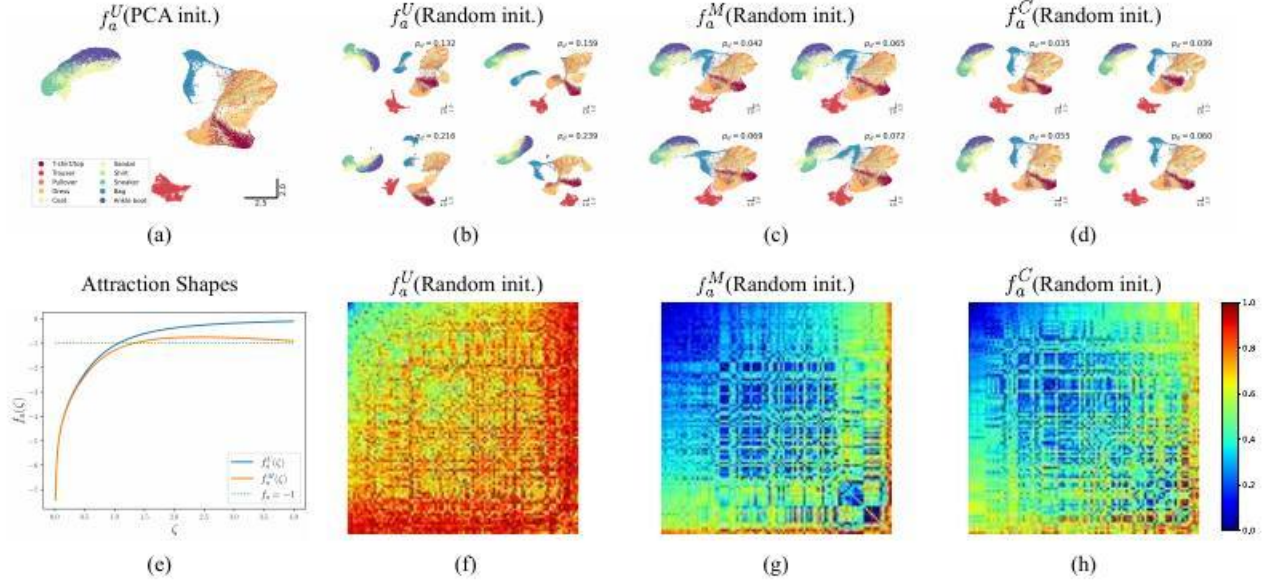


Figure 9: Random-initialization behavior under different attraction shapes on FMNIST. (a) PCA-initialized reference mapping. (b-d) Four mappings with the lowest Procrustes distance (p_d) from (a) for (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices from 100 runs for (f) default (0.72 ± 0.15), (g) modified (0.38 ± 0.19), and (h) composite (0.43 ± 0.18) attraction shapes.

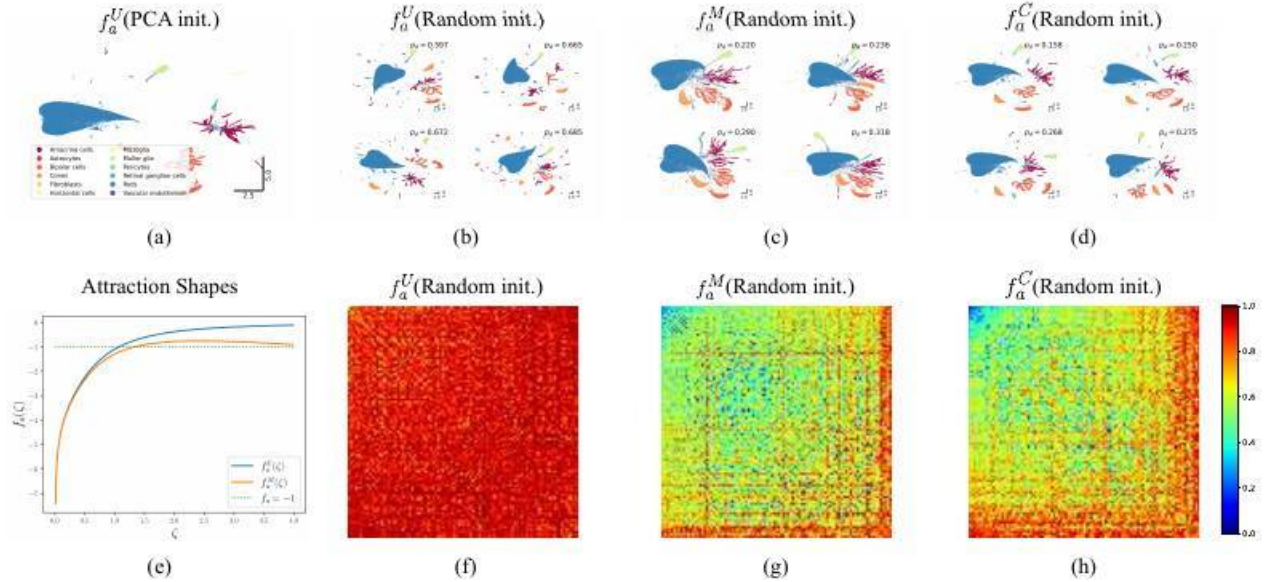


Figure 10: Random-initialization behavior under different attraction shapes on single-cell transcriptomes. (a) PCA-initialized reference mapping. (b-d) Four mappings with the lowest Procrustes distance (p_d) from (a) for (b) default, (c) modified, and (d) composite attraction shapes. (e) Default UMAP and modified attraction shapes. (f-h) Procrustes matrices from 100 runs for (f) default (0.91 ± 0.06), (g) modified (0.61 ± 0.13), and (h) composite (0.64 ± 0.15) attraction shapes.

F More on NEG- t -SNE

F.1 Comparison to UMAP

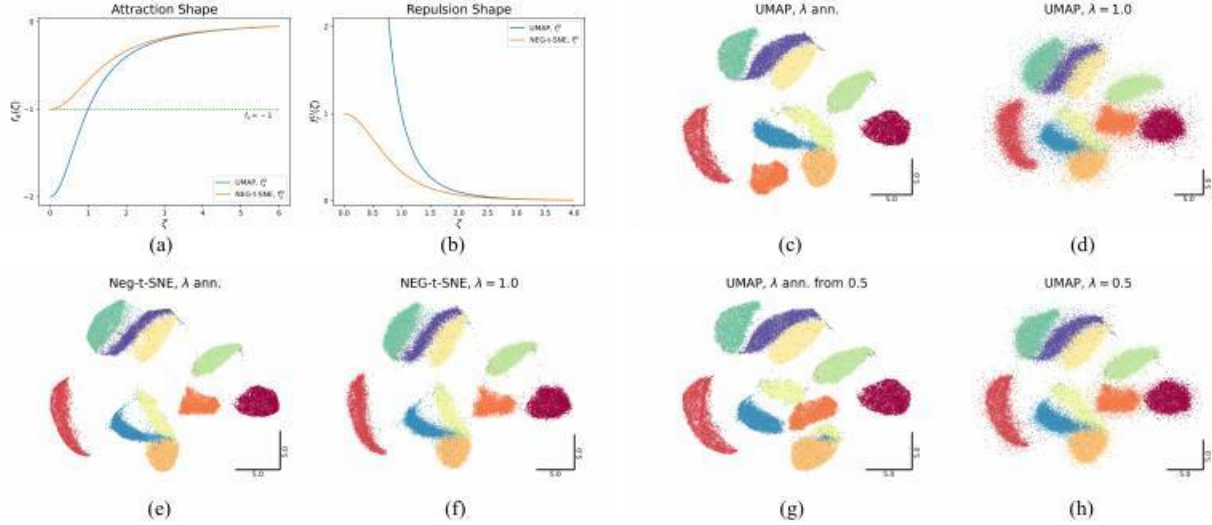


Figure 11: Learning-rate sensitivity of UMAP and NEG- t -SNE on MNIST. (a) Attraction and (b) repulsion shapes for UMAP ($a = 1$, $b = 1$) and NEG- t -SNE. (c,d) UMAP is sensitive to λ because $f_a^U < -1$ at small separation distances ζ ; without annealing, clusters become fuzzy. (e,f) NEG- t -SNE is less sensitive to λ because $f_a^N \in [-1, 0]$, so clusters remain less fuzzy without annealing. (g,h) Confining f_a^U to $[-1, 0]$ by setting $\lambda = 0.5$ reduces UMAP's sensitivity to λ .

Figure 11 compares UMAP and NEG- t -SNE shapes together with MNIST embeddings. With annealed learning rates, the two methods produce similar outputs (Figs. 11(c,e)). With a constant learning rate of 1, however, UMAP produces a fuzzier structure, whereas NEG- t -SNE retains cleaner boundaries (Fig. 11(d,f)). Section 6 suggests that constraining f_a^U to $[-1, 0]$ should reduce fuzziness at fixed λ . The same pattern appears in Fig. 11(h,i), where UMAP gives better embeddings and clustering when the attraction learning rate (λ_a) is $\lesssim 0.5$. A direct way to enforce this condition is to initialize λ at 0.5, satisfying Proposition 4.1 for all ζ . The embeddings in Figs. 11(g) and (h) confirm the prediction: at constant $\lambda = 0.5$, UMAP clusters resemble those of NEG- t -SNE and have sharper boundaries than at $\lambda = 1$ (Fig. 11(d,h)). A few points remain outside clusters because of UMAP's repulsion shape, a behavior reduced by NEG- t -SNE.

The same comparison can be extended by introducing a and b into NEG- t -SNE, which is essentially the Parametric UMAP formulation in Proposition 6.1. The affinity becomes $q_{ij}^N = 1/(2 + a\zeta^{2b})$, and the attraction and repulsion shapes are $f_a^N = -\frac{2ab\zeta^{2(b-1)}}{2+a\zeta^{2b}}$ and $f_r^N = \frac{2ab\zeta^{2(b-1)}}{(1+a\zeta^{2b})(2+a\zeta^{2b})}$, respectively. For $0 < b < 1$, both shapes become unbounded as $\zeta \rightarrow 0$. Thus, varying a and b gives NEG- t -SNE numerical challenges similar to UMAP. A remaining distinction is that NEG- t -SNE reaches a smaller attraction minimum distance (ζ_{-1}) than UMAP; this can support cluster formation, but the attraction also decays faster at longer distances, increasing near-sightedness.

F.2 Comparison to Parametric UMAP

Parametric UMAP was initially trained with the original UMAP objective (Sainburg et al., 2021), but later work adopted a numerically stable modified cross-entropy loss based on a log-sigmoid kernel (Shi et al., 2023). Under this objective, Parametric UMAP and Neg- t -SNE (Damrich et al., 2023) are equivalent (Proposition 6.1). The derivation below makes the correspondence explicit.

Proof of Proposition 6.1

Proof. The kernel function under this modification becomes

$$q_{ij}^P = -\log(1 + a\|y_i - y_j\|_2^{2b}). \quad (35)$$

The attractive term is

$$-\text{logsigmoid}(q_{ij}^P) = -\log\left(\frac{1}{1 + \exp(-q_{ij}^P)}\right) \quad (36)$$

$$= -\log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (37)$$

$$= -\log(q_{ij}^N). \quad (38)$$

And the repulsive term is

$$\text{logsidmoid}(q_{ij}^P) - q_{ij}^P = \log\left(\frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) + \log(1 + a\|y_i - y_j\|_2^{2b}) \quad (39)$$

$$= \log\left(\frac{1 + a\|y_i - y_j\|_2^{2b}}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (40)$$

$$= \log\left(1 - \frac{1}{2 + a\|y_i - y_j\|_2^{2b}}\right) \quad (41)$$

$$= \log(1 - q_{ij}^N). \quad (42)$$

Together, these terms recover NEG-t-SNE with explicit parameters a and b ; in the default Neg- t -SNE formulation, both are set to 1. $\square$

G Alternate Dimensionality Reduction Algorithms

The additional algorithms considered here use the same low-dimensional Cauchy kernel as UMAP with $a = 1$ and $b = 1$:

$$q_{ij} = \frac{1}{1 + \|y_i - y_j\|_2^2}. \quad (43)$$

We begin with TriMap (Amid and Warmuth, 2019). Although TriMap uses triplets rather than pairwise interactions, it provides a useful bridge from UMAP’s separable forces to more coupled attraction-repulsion behavior. We then discuss Pairwise Controlled Manifold Approximation (PaCMAP) (Wang et al., 2021) and its extension, Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP) (Wang et al., 2025), both of which adapt TriMap-style ideas to pairwise losses. Finally, we examine t -SNE (Van der Maaten and Hinton, 2008), briefly discuss SNE (Hinton and Roweis, 2002) with its alternate kernel, and close with multidimensional scaling (Borg and Groenen, 2007).

G.1 TriMap

TriMap (Amid and Warmuth, 2019) optimizes the low-dimensional embedding with a triplet loss

$$\mathcal{L}^T = \sum_{(i,j,k)} w_{ijk} \frac{1}{1 + \frac{q_{ij}}{q_{ik}}}, \quad (44)$$

where w_{ijk} weights the triplet (y_i, y_j, y_k) , y_j belongs to the high-dimensional k -nearest-neighbor set of y_i , and y_k is a far-away point. Minimizing this loss should attract y_i and y_j while repelling y_i and y_k . The corresponding updates are

$$y_i^{t+1} = y_i^t + \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t) + \lambda f_r^T(\zeta_1, \zeta_2)(y_i^t - y_k^t), \quad (45)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_j^t), \quad (46)$$

$$y_k^{t+1} = y_k^t - \lambda f_r^T(\zeta_1^t, \zeta_2^t)(y_i^t - y_k^t), \quad (47)$$

Here, $\zeta_1 = \|y_i - y_j\|_2$ is the nearest-neighbor distance, $\zeta_2 = \|y_i - y_k\|_2$ is the distance to the far-away point, and f_a^T and f_r^T are TriMap’s attraction and repulsion shapes. Unlike UMAP’s pairwise case, TriMap’s attractive and repulsive components are coupled and depend on two distances, making the shapes two-dimensional. The attraction shape is

$$f_a^T(\zeta_1, \zeta_2) = -\frac{2(1 + \zeta_2^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}, \quad (48)$$

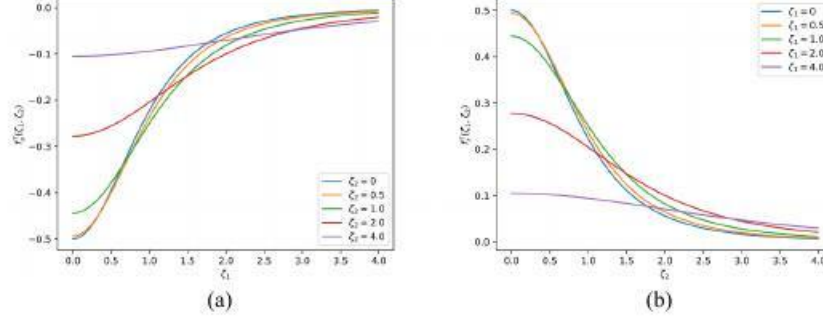


Figure 12: TriMap shapes under coupled triplet distances. (a) Attraction shapes for different ζ_2 . (b) Repulsion shapes for different ζ_1 .

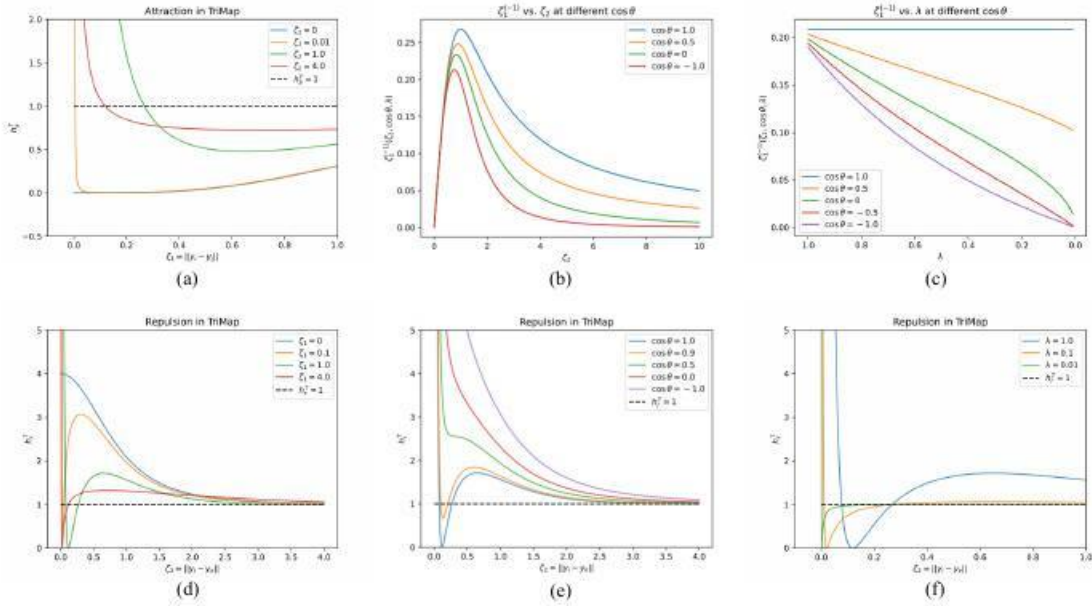


Figure 13: Attraction and repulsion behavior in TriMap. (a) h_a^T versus ζ_1 for different ζ_2 ; values below the dotted line indicate attraction. As in UMAP, nearest-neighbor pairs (y_i, y_j) can experience both attraction and repulsion. (b,c) The minimum distance for contraction (c_1^{-1}) varies with (b) ζ_2 and (c) λ , with $\cos \theta$ regulating the range of values. (d-f) Repulsion in TriMap when varying (d) ζ_1 , (e) $\cos \theta$, and (f) λ . Values above (below) the dotted line indicate repulsion (attraction). Unlike UMAP's pairwise repulsion, TriMap's repulsive component can induce either expansion or contraction. Unless otherwise labeled, $\zeta_1 = 1.0$, $\zeta_2 = 0.5$, $\cos \theta = 1.0$, and $\lambda = 1.0$.

and the repulsion shape is

$$f_r^T(\zeta_1, \zeta_2) = \frac{2(1 + \zeta_1^2)}{(2 + \zeta_1^2 + \zeta_2^2)^2}. \quad (49)$$

TriMap's attraction and repulsion shapes (Fig. 12) show trends similar to UMAP's, but the minimum attraction value is -0.5 ; unlike UMAP, attraction alone therefore does not flip point positions. Because Eqs. (45)-(47) couple attraction and repulsion, Propositions 4.1 and 4.2 do not apply directly. We first characterize contraction:

Proposition G.1. *Update equations (45)-(47) provide a contraction if*

$$h_a^T(\zeta_1, \zeta_2, \theta, \lambda) < 1, \quad (50)$$

where $h_a^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_a^T)^2 + 2(1 + 2\lambda f_a^T)\lambda f_r^T \frac{\zeta_2}{\zeta_1} \cos \theta + (\lambda f_r^T)^2 \frac{\zeta_2^2}{\zeta_1^2}$ and θ is the angle between the vectors $(y_i - y_j)$ and $(y_i - y_k)$, i.e., $\cos \theta = \frac{(y_i^t - y_j^t)^T (y_i^t - y_k^t)}{\|y_i^t - y_j^t\|_2 \|y_i^t - y_k^t\|_2}$.

Proof. We require

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 < \|y_i^t - y_j^t\|_2^2. \quad (51)$$

From Eq. (45) and (46): $y_i^{t+1} - y_j^{t+1} = (1 + 2\lambda f_a^T)(y_i^t - y_j^t) + \lambda f_r^T(y_i^t - y_k^t)$. Putting this value in Eq. (51), we obtain the desired inequality. $\square$

Inequality (50) depends on ζ_1 , ζ_2 , $\cos \theta$, and λ , so contraction of the nearest-neighbor distance $\zeta_1 = \|y_i - y_j\|_2$ is coupled to additional triplet geometry. Figure 13(a) shows attraction behavior for different ζ_2 values with $\cos \theta = 1$ and $\lambda = 1$. Values below the dotted line indicate attraction and contraction of ζ_1 , whereas values above indicate repulsion and expansion. The crossing with the dotted line defines the minimum distance for contraction ($\zeta_1^{(-1)}$), analogous to UMAP's ζ_{-1} :

$$\zeta_1^{(-1)}(\zeta_2, \theta, \lambda) = \arg \min_{\zeta_1} |h_a^T(\zeta_1, \zeta_2, \theta, \lambda) - 1|, \quad (52)$$

$$\text{s.t. } \zeta_1 \geq 0. \quad (53)$$

$\zeta_1^{(-1)}$ is finite and positive in most cases (Fig. 13(b,c)). Thus, despite its triplet formulation, TriMap can exhibit a UMAP-like need for learning-rate adaptation or an analogous mechanism; the TriMap implementation uses delta-bar-delta (Jacobs 1988) under appropriate initialization.

Proposition G.2. *Update equations (45)-(47) provide expansion if*

$$h_r^T(\zeta_1, \zeta_2, \theta, \lambda) > 1, \quad (54)$$

where $h_r^T(\zeta_1, \zeta_2, \theta, \lambda) = (1 + 2\lambda f_r^T)^2 + 2\lambda f_a^T(1 + \lambda f_r^T) \frac{\zeta_1}{\zeta_2} \cos \theta + (\lambda f_a^T)^2 \frac{\zeta_1^2}{\zeta_2^2}$.

Proof. We require

$$\|y_i^{t+1} - y_k^{t+1}\|_2^2 > \|y_i^t - y_k^t\|_2^2. \quad (55)$$

From Eq. (45) and (47): $y_i^{t+1} - y_k^{t+1} = (1 + 2\lambda f_r^T)(y_i^t - y_k^t) + \lambda f_a^T(y_i^t - y_j^t)$. Putting this value in Eq. (55) we obtain inequality (54). $\square$

Inequality (54) also depends on ζ_1 , ζ_2 , $\cos \theta$, and λ . Here the target behavior is expansion of $\zeta_2 = \|y_i - y_k\|_2$. Figures 13(d-f) vary the other quantities to show when the update is repulsive. Values above the dotted line indicate expansion, whereas values below indicate attraction. The key contrast with UMAP is that TriMap's repulsive component can sometimes contract distances. Because this occurs at small distances, appropriate initialization and triplet selection can avoid the problematic regime.

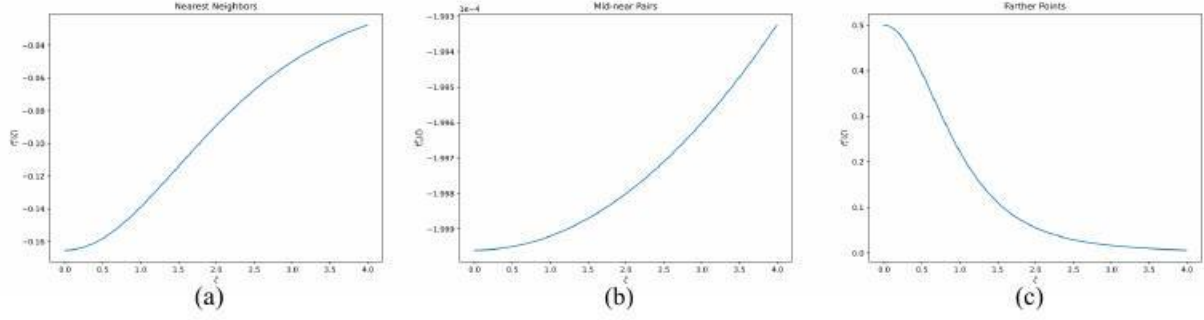


Figure 14: PaCMAP attraction and repulsion shapes. (a,b) Attraction shapes for (a) nearest-neighbor and (b) mid-near pairs; mid-near values are on the order of 10^{-4} . (c) Repulsion shape for farther pairs. $\lambda = 1$ in all panels.

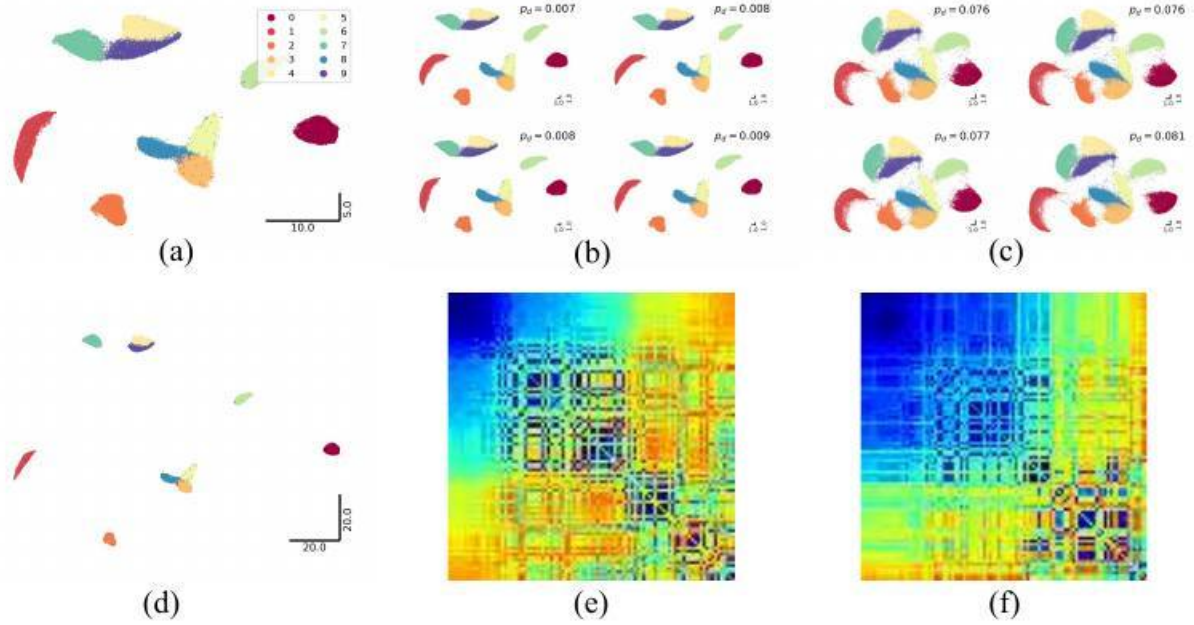


Figure 15: PaCMAP behavior under initialization and shape changes. (a) PCA-initialized PaCMAP embedding of MNIST. (b,c) Four random-initialized samples closest to (a) for (b) original PaCMAP and (c) PaCMAP with modified attraction $f_a^M(\zeta) = f_a^P(\zeta) - 0.001\zeta$. (d) PaCMAP on MNIST with modified repulsion $f_r^M = f_r^P + 0.00005$. (e,f) Procrustes matrices for (e) original PaCMAP (0.51 ± 0.22) and (f) PaCMAP with modified attraction (0.43 ± 0.22). As in UMAP, stronger long-distance attraction improves consistency, while stronger repulsion yields smaller clusters and larger inter-cluster distances.

G.2 Pairwise Controlled Manifold Approximation (PaCMAP)

PaCMAP (Wang et al., 2021) optimizes the low-dimensional embedding across multiple pair scales. Its loss is

$$\mathcal{L}^P = w_{NB} \sum_{(i,j) \in NN} \frac{1}{1 + 10q_{ij}} + w_{MN} \sum_{(i,k) \in MN} \frac{1}{1 + 10000q_{ik}} + w_{FP} \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}, \quad (56)$$

where w_{NB} , w_{MN} , and w_{FP} weight nearest-neighbor (NN) pairs, mid-near (MN) pairs, and further pairs (FP), respectively (details in Appendix H). The first two terms are attractive, while the final term is repulsive. From the shape perspective, this loss is a pairwise, separable modification of TriMap’s triplet loss: the attractive terms replace TriMap’s far-point affinity with constants ($1/10$ for nearest neighbors, $1/10000$ for mid-near pairs, and 1 for farther points). This

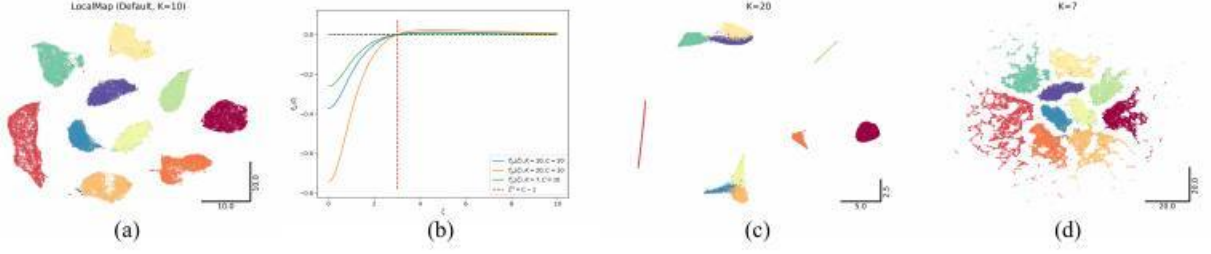


Figure 16: LocalMAP behavior on MNIST. (a) Default embedding. (b) Nearest-neighbor attraction-repulsion shape; values below (above) the dotted line indicate attraction (repulsion). The transition occurs as ζ crosses $\sqrt{C-1}$, with $C = 10$ as in the LocalMAP implementation. (c) With large K ($= 20$), repulsion dominates and clusters become compact. (d) With small K ($= 7$), attraction dominates and clusters break apart.

separability decouples the update equations. The nearest-neighbor updates are

$$y_i^{t+1} = y_i^t + \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (57)$$

$$y_j^{t+1} = y_j^t - \lambda f_a^P(\zeta_1^t)(y_i^t - y_j^t), \quad (58)$$

the mid-near-pair updates are

$$y_i^{t+1} = y_i^t + \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (59)$$

$$y_k^{t+1} = y_k^t - \lambda f_m^P(\zeta_2^t)(y_i^t - y_k^t), \quad (60)$$

and the farthest-pair updates are

$$y_i^{t+1} = y_i^t + \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (61)$$

$$y_l^{t+1} = y_l^t - \lambda f_r^P(\zeta_3^t)(y_i^t - y_l^t), \quad (62)$$

where $\zeta_1 = \|y_i - y_j\|_2$, $\zeta_2 = \|y_i - y_k\|_2$, and $\zeta_3 = \|y_i - y_l\|_2$ are the relevant pair distances. The functions f_a^P and f_m^P are attraction shapes for nearest neighbors and mid-near pairs, respectively, and f_r^P is the repulsion shape for farthest pairs. Their forms are

$$f_a^P(\zeta) = -\frac{20}{(11 + \zeta^2)^2}, \quad (63)$$

$$f_m^P(\zeta) = -\frac{20000}{(10001 + \zeta^2)^2}, \quad (64)$$

$$f_r^P(\zeta) = \frac{2}{(2 + \zeta^2)^2}. \quad (65)$$

f_a^P and f_m^P satisfy Proposition 4.1 and f_r satisfies Proposition 4.2 (Fig. 14). PaCMAP’s attraction is weaker than UMAP’s, but it remains effective across a wide range of learning rates when modulated by Adam (Kingma and Ba 2015); with $w_{NB} = 3$, the maximum attraction stays below 0.5, preventing flips during attractive updates. PaCMAP typically initializes the embedding inside a small low-dimensional sphere (e.g., on the order of 10^{-3} in (Wang et al. 2021)) and then relies on repulsion to separate clusters. In these experiments, it largely recovers UMAP-like clustering properties, especially on MNIST, while improving ordering. Its random-initialization consistency is better than UMAP’s (Fig. 15 b,e) and improves further with modified attraction (Fig. 15 c,f). Adding a small value to the repulsion shape increases embedding compactness by increasing inter-cluster distance.

G.3 Pairwise Controlled Manifold Approximation with Local Adjusted Graph (LocalMAP)

LocalMAP (Wang et al. 2025) extends PaCMAP. In the MNIST experiment, one of its defining behaviors is separation of all 10 clusters (Fig. 16 a), similar to the patterns in Figs. 3 g,h,j,k and Figs. 4 a,b). We use it as another example of how attractive and repulsive shapes jointly control cluster compactness and connectedness.

LocalMAP first performs PaCMAP and then applies additional optimization designed to decouple clusters. It minimizes

$$\mathcal{L}^L = \sum_{(i,j) \in NN} \frac{K}{\frac{1}{\sqrt{q_{ij}}} + C\sqrt{q_{ij}}} + \sum_{(i,l) \in FP} \frac{1}{1 + \frac{1}{q_{il}}}. \quad (66)$$

The first term combines attractive and repulsive behavior on the same nearest-neighbor pair. In one distance regime it attracts, while in another it repels. The second term matches the PaCMAP far-pair term, except that LocalMAP resamples farther pairs every few iterations. We therefore focus on the first term involving nearest neighbors. The updates are

$$y_i^{t+1} = y_i^t + \lambda f_{ar}^L(\zeta_i^t)(y_i^t - y_j^t), \quad (67)$$

$$y_j^{t+1} = y_j^t - \lambda f_{ar}^L(\zeta_j^t)(y_i^t - y_j^t), \quad (68)$$

where f_{ar}^L is the attraction-repulsion shape given by

$$f_{ar}^L(\zeta) = -\frac{K(C-1-\zeta^2)}{2\sqrt{1+\zeta^2}(1+C+\zeta^2)^2}. \quad (69)$$

The update contracts when $\zeta^2 < C-1$ and $-1 < \lambda f_{ar}^L < 0$ by Proposition 4.1. When $\zeta^2 > C-1$, $f_{ar} > 0$, so Proposition 4.2 implies expansion. Thus K and C determine whether attraction or repulsion dominates. In the LocalMAP implementation, both C and K are set to 10 (Fig. 16(b)), and the attraction is stronger than in PaCMAP (Fig. 14(a)). Consequently, when $\zeta > 3$, nearest-neighbor pairs are repelled, causing bridge pairs between clusters to separate. The parameter K scales the forces and regulates this separation.

By Proposition 4.1 requiring $\lambda f_{ar}^L(0) \geq -\frac{1}{2}$ gives the largest K values, and hence the strongest f_{ar}^L attraction, that avoid flipping pair placements. Avoiding flips helps preserve the PaCMAP ordering before the LocalMAP refinement; otherwise, cluster separation can be inhibited. Equation 69 then simplifies to $K \leq \frac{\lambda(1+C)^2}{C-1}$, which gives $K \lesssim 13.44$ for $C = 10$ and $\lambda = 1$. For $K \gtrsim 13.44$, repulsion dominates and clusters become more compact, but the LocalMAP objective is not achieved because bridge structures persist (Fig. 16(c) for $K = 20$). As K decreases, attraction dominates because repulsion is too weak, and the embedding breaks existing clusters apart (Fig. 16(d) for $K = 7$), matching the behavior in Fig. 4(b).

G.4 t -distributed Stochastic Neighbor Embedding (t -SNE)

t -SNE (Van der Maaten and Hinton, 2008) also optimizes pairwise relationships, but with a globally normalized objective:

$$\mathcal{L} = -\sum_{i,j} w_{i,j} \log \left(\frac{q_{i,j}}{\sum_{k \neq l} q_{k,l}} \right), \quad (70)$$

where $w_{i,j}$ is the pair weight. The original t -SNE formulation considers all pairs, not only nearest neighbors. The loss decomposes into attraction and repulsion as

$$\mathcal{L} = \sum_{i,j} \left[-w_{i,j} \log(q_{i,j}) + w_{i,j} \log \left(\sum_{k \neq l} q_{k,l} \right) \right]. \quad (71)$$

As before, the first term is attractive and the second is repulsive. The attractive term matches UMAP with $a = 1$ and $b = 1$ and is straightforward to compute, whereas the repulsive term couples all pairs and is therefore expensive. Applying the same shape decomposition used for UMAP gives the t -SNE updates. Because the original t -SNE formulation does not rely on a nearest-neighbor graph, the all-pairs weights $w_{i,j}$ remain central in the updates. The attractive updates are

$$y_i^{t+1} = y_i^t + \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (72)$$

$$y_j^{t+1} = y_j^t - \lambda w_{i,j} f_a^{t-SNE}(\zeta_{i,j}^t)(y_i^t - y_j^t), \quad (73)$$

where f_a^{t-SNE} is the attraction shape of t -SNE and $\zeta_{i,j} = \|y_i - y_j\|_2$. The update equation for the repulsive parts are

$$y_i^{t+1} = y_i^t + \lambda \frac{w_{i,j}}{Z} \sum_k f_r^{t-SNE}(\zeta_{i,k}^t)(y_i^t - y_k^t), \quad (74)$$

$$y_j^{t+1} = y_j^t - \lambda \frac{w_{i,j}}{Z} \sum_l f_r^{t-SNE}(\zeta_{l,j}^t)(y_l^t - y_j^t), \forall j, j \neq i, \quad (75)$$

where f_r^{t-SNE} is the t -SNE repulsion shape and $Z = \sum_{k \neq l} q_{k,l}$. The shape functions are

$$f_a^{t-SNE}(\zeta) = -\frac{2}{1 + \zeta^2}, \text{ and} \quad (76)$$

$$f_r^{t-SNE}(\zeta) = \frac{2}{(1 + \zeta^2)^2}. \quad (77)$$

From the attractive updates in Eqs. (72)-(73), f_a^{t-SNE} satisfies Proposition 4.1 when $0 < \lambda w_{i,j} f_a^{t-SNE} < -1$ and defines a minimum distance for contraction, ζ_{-1} . The repulsive update is coupled across all pairs, so it cannot be read directly from the scalar repulsion shape. Instead, using the same coupled-update reasoning as in TriMap, we obtain:

Proposition G.3. *The update Eqs. (74)-(75) provide an expansion if*

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) > 1, \quad (78)$$

where

$$h_r^{t-SNE}(\zeta_{i,j}, v, \theta, \lambda, w_{i,j}) = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))^2 + \frac{\|v\|_2^2}{\zeta_{i,j}^2} + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}) \frac{\|v\|_2}{\zeta_{i,j}} \cos \theta,$$

$$v = \lambda \frac{w_{i,j}}{Z} \left(\sum_{k, k \neq j} f_r^{t-SNE}(\zeta_{i,k})(y_i - y_k) + \sum_{l, l \neq i} f_r^{t-SNE}(\zeta_{l,j})(y_l - y_j) \right),$$

and θ is the angle between $(y_i - y_j)$ and v .

Proof. We require,

$$\|y_i^{t+1} - y_j^{t+1}\|_2^2 > \|y_i^t - y_j^t\|_2^2. \quad (79)$$

From Eqs. (74) and (75), $y_i - y_j = (1 + 2\lambda \frac{w_{i,j}}{Z} f_r^{t-SNE}(\zeta_{i,j}))(y_i - y_j) + v$. Putting this value in Eq. (79), we obtain the desired inequality. $\square$

Because the t -SNE repulsion condition in Eq. (78) has the same coupled form as TriMap's condition, its repulsive update can also induce either expansion or contraction depending on the surrounding geometry.

Because t -SNE forces are scaled by $w_{i,j}$ for attraction and $\frac{w_{i,j}}{Z}$ for repulsion, they are typically smaller than UMAP's corresponding forces. Implementations therefore often use large learning rates, such as $\approx 10^3$ in Open- t -SNE (Poličar et al. 2024). Most t -SNE implementations also use early exaggeration, multiplying attractive forces by a constant during the first iterations. This increases far-sightedness by pulling together points that should be close but begin far apart. At the same time, points that start too close ($\zeta_{i,j} \approx \zeta_{-1}$) but should separate can also be rearranged. Thus, early exaggeration is central to obtaining consistent t -SNE embeddings, especially under random initialization. When applied throughout optimization, it makes t -SNE embeddings resemble UMAP embeddings more closely (Böhm et al. 2022).

G.5 Stochastic Neighbor Embedding (SNE)

SNE (Hinton and Roweis 2002) is one of the earliest algorithms in this family. It uses the same overall form as t -SNE (Eq. 71) but replaces the low-dimensional kernel with $q_{i,j} = \exp(-\|y_i - y_j\|_2^2)$. This gives the attraction shape

$$f_a^{SNE}(\zeta) = -2, \quad (80)$$

that follows Proposition 4.1 (with $0 < \lambda w_{i,j} f_a^{SNE} < -1$) and the repulsion shape

$$f_r^{SNE}(\zeta) = 2 \exp(-\zeta^2), \quad (81)$$

which follows Proposition G.3 after replacing the t -SNE symbols with their SNE counterparts. The constant attraction shape makes contraction depend heavily on the learning rate (λ) and pair weights ($w_{i,j}$), leading to overlap even for small samples—the crowding problem (Van der Maaten and Hinton 2008). t -SNE and later algorithms reduce this issue by moving from a Gaussian kernel to a heavy-tailed kernel such as Eq. (43).

G.6 Multidimensional Scaling (MDS)

Multidimensional scaling (MDS) (Borg and Groenen 2007) is usually optimized by stress majorization rather than gradient descent, since naive gradient methods often fail to converge to useful mappings. Still, gradient formulations have been studied (Kruskal 1964; Zheng et al. 2018), and they provide another example of attraction-repulsion behavior. In particular, Zheng et al. (2018) proposes a successful gradient-descent MDS method for graph drawing. We begin

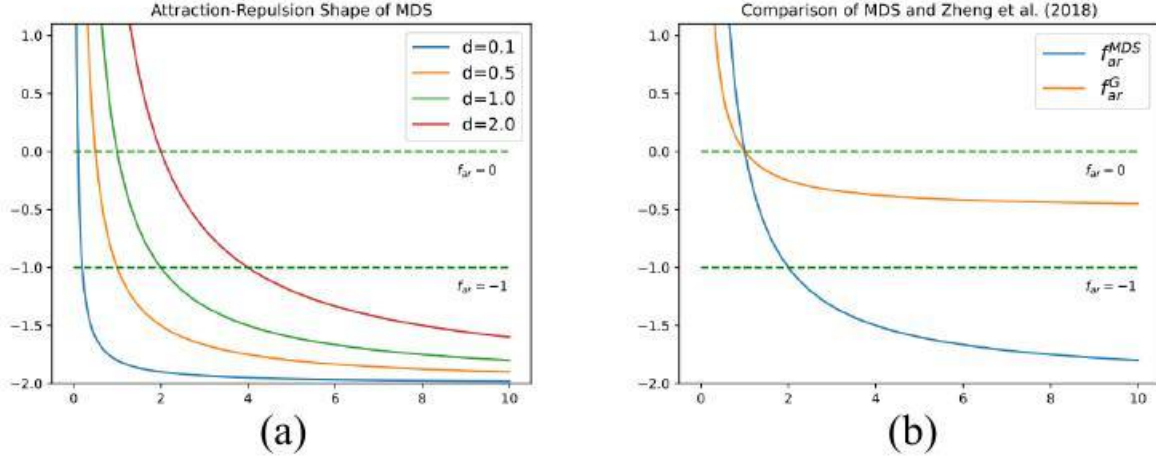


Figure 17: Attraction-repulsion behavior in MDS. (a) MDS characteristic using f_{ar} . (b) Comparison between the MDS characteristic and the method in [Zheng et al. \(2018\)](#) for $d = 1.0$.

with the cost function, ignoring $w_{i,j}$ without loss of generality for this discussion because it can be absorbed into the learning rate:

$$\mathcal{L}^{MDS} = \sum_{i,j,i < j} (\|y_i - y_j\| - d_{ij})^2. \quad (82)$$

The loss contains one term per pair rather than separate attraction and repulsion terms, but each term can act attractively or repulsively depending on the current distance. Because the loss separates over individual (i, j) components, we can analyze a single pair. The derivative is

$$\frac{\partial}{\partial y_i} (\|y_i - y_j\| - d_{ij})^2 = 2 \frac{\|y_i - y_j\| - d_{ij}}{\|y_i - y_j\|} (y_i - y_j). \quad (83)$$

Thus, the attraction-repulsion shape of MDS becomes

$$f_{ar}^{MDS}(\zeta) = -2 \frac{\zeta - d}{\zeta}, \quad (84)$$

where $\zeta = \|y_i - y_j\|_2$. The shape is attractive when $\zeta > d$ and is bounded within $[-2, 0]$ in that region (Fig. 17(a)). This range is not always suitable for convergence, because only values in $[-1, 0]$ contract while the others expand; values below -2 cause point flips followed by expansion. To avoid this regime, [Zheng et al. \(2018\)](#) uses a force-directed-graph-inspired formulation:

$$f_{ar}^G(\zeta) = -\frac{1}{2} \frac{\zeta - d}{\zeta}, \quad (85)$$

This shape works with effective learning rate $\lambda \leq 1$ because f_{ar}^G is bounded in $[-0.5, 0]$. For $\zeta < d$, $f_{ar}^{MDS} > 0$ produces repulsion. Overall, this attraction-repulsion interaction is most effective when the initialization is already near a desired output. Continued optimization of a single pair oscillates around $\zeta = d$, where $f_{ar}^G = 0$, yielding a distance threshold analogous to UMAP's ζ_{-1} . Unlike UMAP, however, no learning-rate choice drives this threshold to zero; preserving it is the objective of MDS. As a result, clusters are often fuzzier than in methods such as UMAP and PaCMAP (for illustrations, see [Lambert et al. \(2022\)](#), [de Bodt et al. \(2025\)](#), and [Kury et al. \(2025\)](#)). [Lambert et al. \(2022\)](#) addresses related issues by converting the MDS loss to quartet stress, involving four samples, and using relative distances, defined by dividing a distance by the sum of six quartet distances; this enables global regularization and lower computation than majorization.

H Construction of the High-Dimensional Graph

Stochastic Neighbor Embedding (SNE) [Hinton and Roweis \(2002\)](#) underpins many modern dimensionality reduction algorithms. It constructs a high-dimensional graph for the dataset $X = \{x_i \in R^n | i = 1, \dots, N\}$ using

$$w_{ij} = \frac{p_{j|i} + p_{i|j}}{2N}, \quad (86)$$

$$w_{j|i} = \frac{\exp(-\|x_i - x_j\|_2^2 / 2\sigma_i^2)}{\sum_{t \neq v} \exp(-\|x_t - x_v\|_2^2 / 2\sigma_t^2)}, \quad (87)$$

where σ_i^2 is chosen to match the user-defined *perplexity* P , defined by

$$P = 2^{H_i} \quad (88)$$

$$H_i = \sum_{j \neq i} w_{j|i} \log_2 w_{j|i}. \quad (89)$$

UMAP instead builds its high-dimensional graph from a k-nearest-neighbor (k-NN) construction:

$$p_{i|j} = \begin{cases} \exp\left(-\frac{d(x_i, x_j) - \rho_i}{\sigma_i}\right) & \text{if } x_j \in \text{KNN}(x_i, k) \\ 0 & \text{otherwise} \end{cases}, \quad (90)$$

$$\rho_i = \min_{x_j \in \text{KNN}(x_i, k)} d(x_i, x_j), \quad (91)$$

where $\text{KNN}(x_i, k)$ is the set of k nearest neighbors of x_i , and σ_i is chosen so that $\sum_j p_{i|j} = \log_2(k)$. The graph is then symmetrized by a t-conorm:

$$p_{i,j} = p_{i|j} + p_{j|i} - p_{i|j}p_{j|i}. \quad (92)$$

PaCMAP uses the k-NN graph with affinities all set to 1, together with a self-tuning distance measure (Zelnik-Manor and Perona 2004),

$$d_{i,j}^2 = \frac{\|x_i - x_j\|^2}{\sigma_i \sigma_j}, \quad (93)$$

where σ_i is the average distance from x_i to its fourth through sixth Euclidean nearest neighbors. This parameter plays a role analogous to the σ_i terms in Eq. (87) and Eq. (90), although the definitions differ.

These graph-construction choices determine the high-dimensional affinities, but our analysis indicates that the subsequent optimization—especially the attraction and repulsion shapes—is the primary driver of the final low-dimensional embedding.

I Detailed Results for Varying λ_a and λ_b

This section provides detailed results for the sweeps summarized in Fig. II h,i). We generate these results by replacing UMAP’s attraction and repulsion shapes with those of other methods while keeping the evaluation protocol fixed. Figure 18 reproduces the main-text summary. Figures 19-22 show the individual MNIST embeddings for each choice of λ_a and λ_r , together with the initialization, an annealed-learning-rate reference embedding, and cases where either λ_a or λ_r is set to 0. Because PaCMAP also uses mid-near pairs, we include an additional reference with mid-near pairs enabled. Figures 23-27 and Figures 28-32 repeat the same sweep logic on FMNIST and single-cell transcriptomes, respectively.

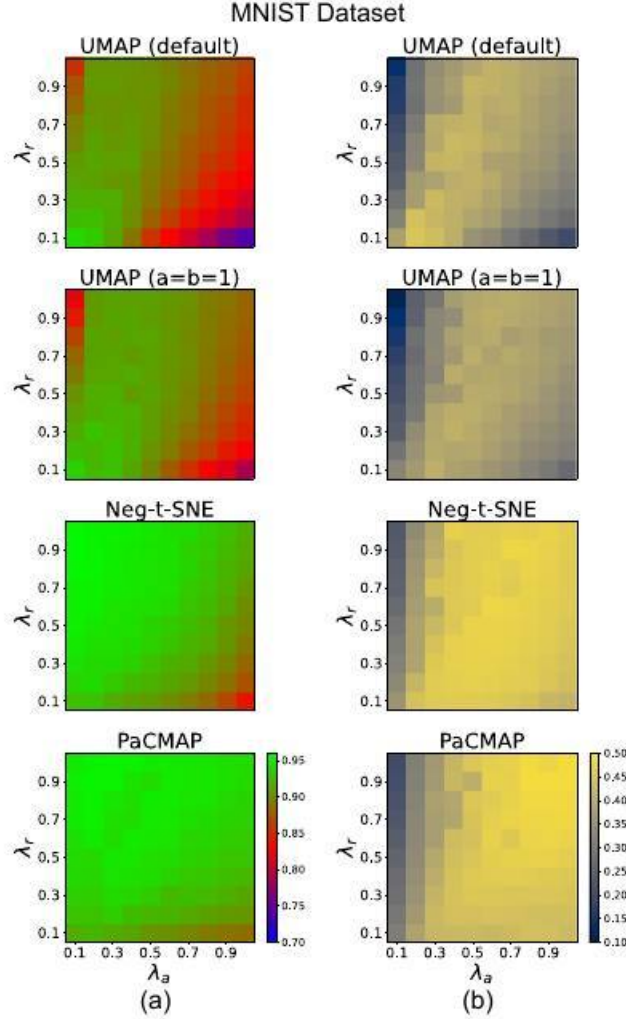


Figure 18: MNIST summary metrics for the learning-rate sweeps: (a) trustworthiness and (b) silhouette score for the compared methods.

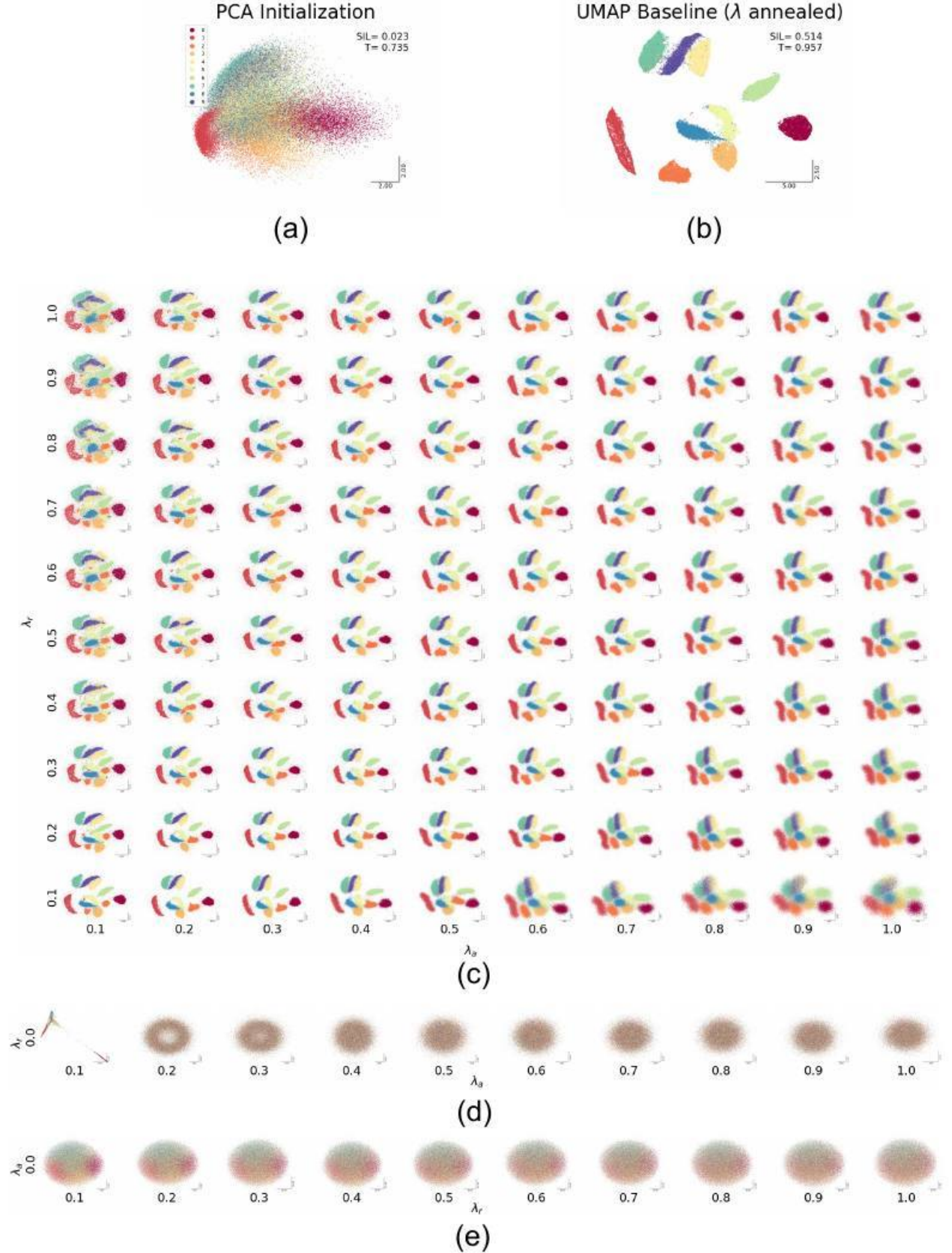


Figure 19: MNIST embeddings while varying λ_a and λ_r using UMAP's attraction and repulsion shapes. (a) Initialization used for the embeddings. (b) Annealed-learning-rate baseline with λ starting from 1. (c) Embeddings obtained by varying λ_a and λ_r without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone does not produce clusters. (e) With $\lambda_a = 0$ (no attraction), repulsion alone also does not produce clusters.

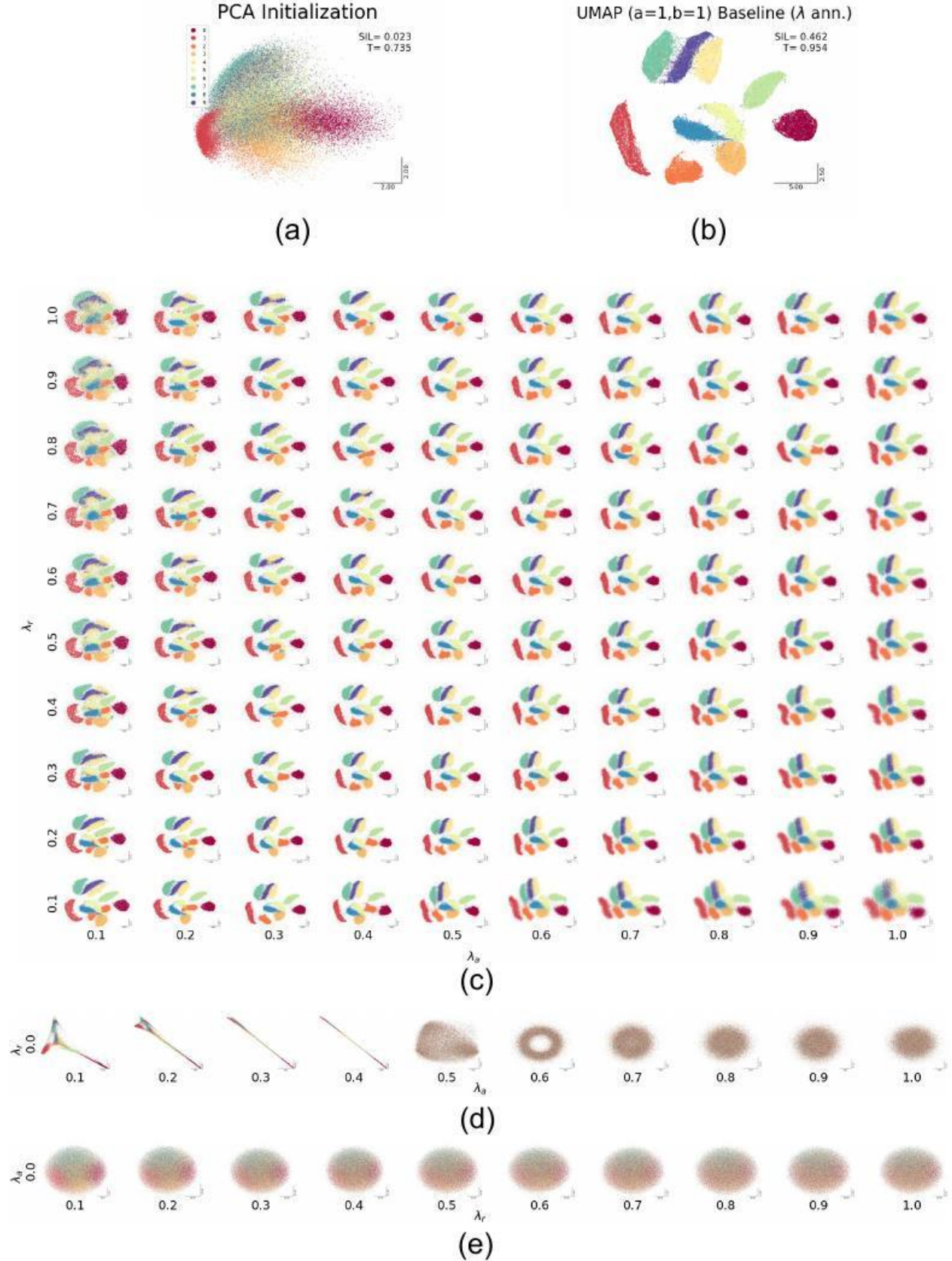


Figure 20: MNIST embeddings while varying λ_a and λ_r using UMAP's attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Initialization used for the embeddings. (b) Annealed-learning-rate baseline with λ starting from 1. (c) Embeddings obtained by varying λ_a and λ_r without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone does not produce clusters. (e) With $\lambda_a = 0$ (no attraction), repulsion alone also does not produce clusters.

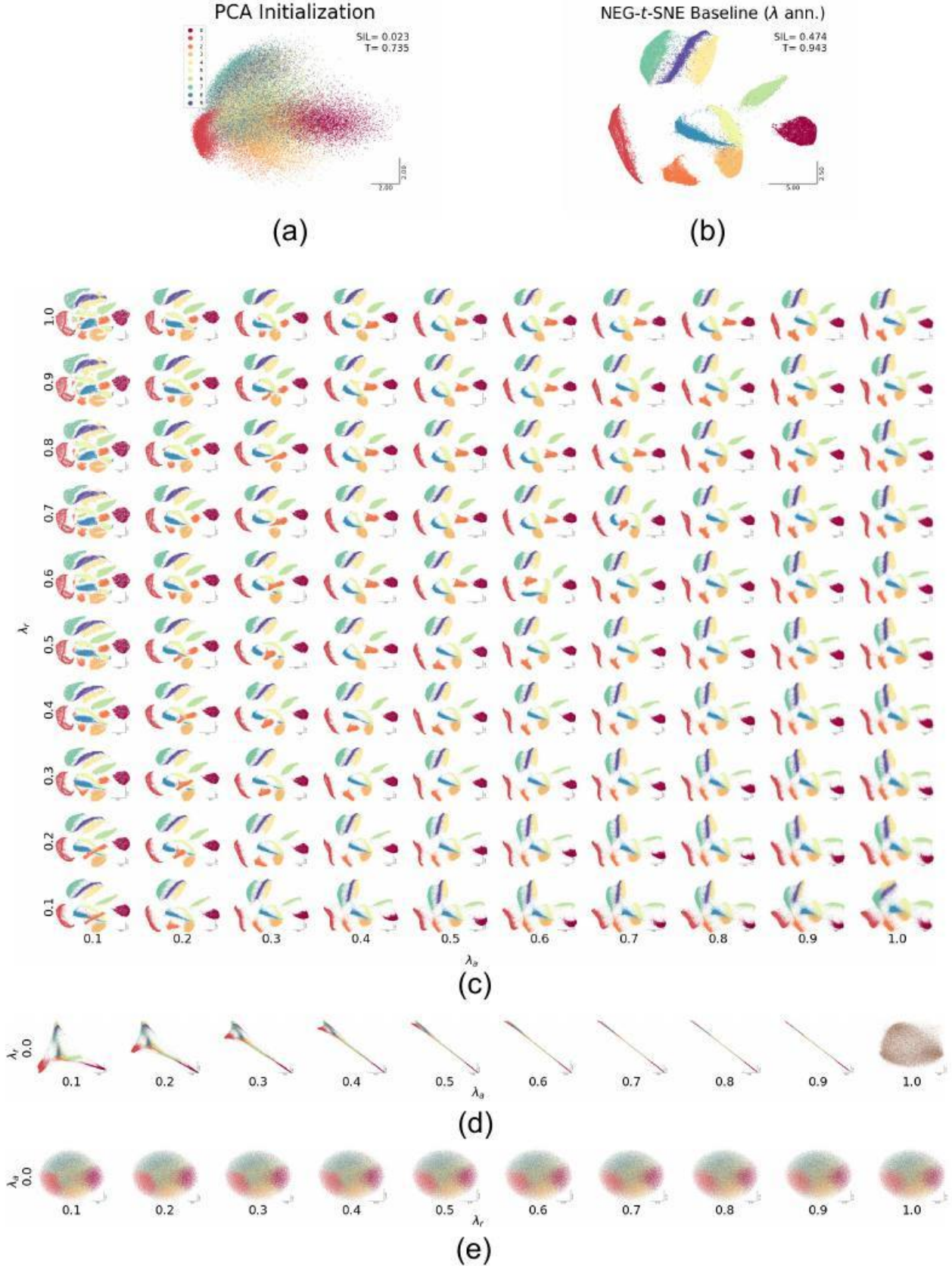


Figure 21: MNIST embeddings while varying λ_a and λ_r using NEG-t-SNE's attraction and repulsion shapes. (a) Initialization used for the embeddings. (b) Annealed-learning-rate baseline with λ starting from 1. (c) Embeddings obtained by varying λ_a and λ_r without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone does not produce clusters. (e) With $\lambda_a = 0$ (no attraction), repulsion alone also does not produce clusters.

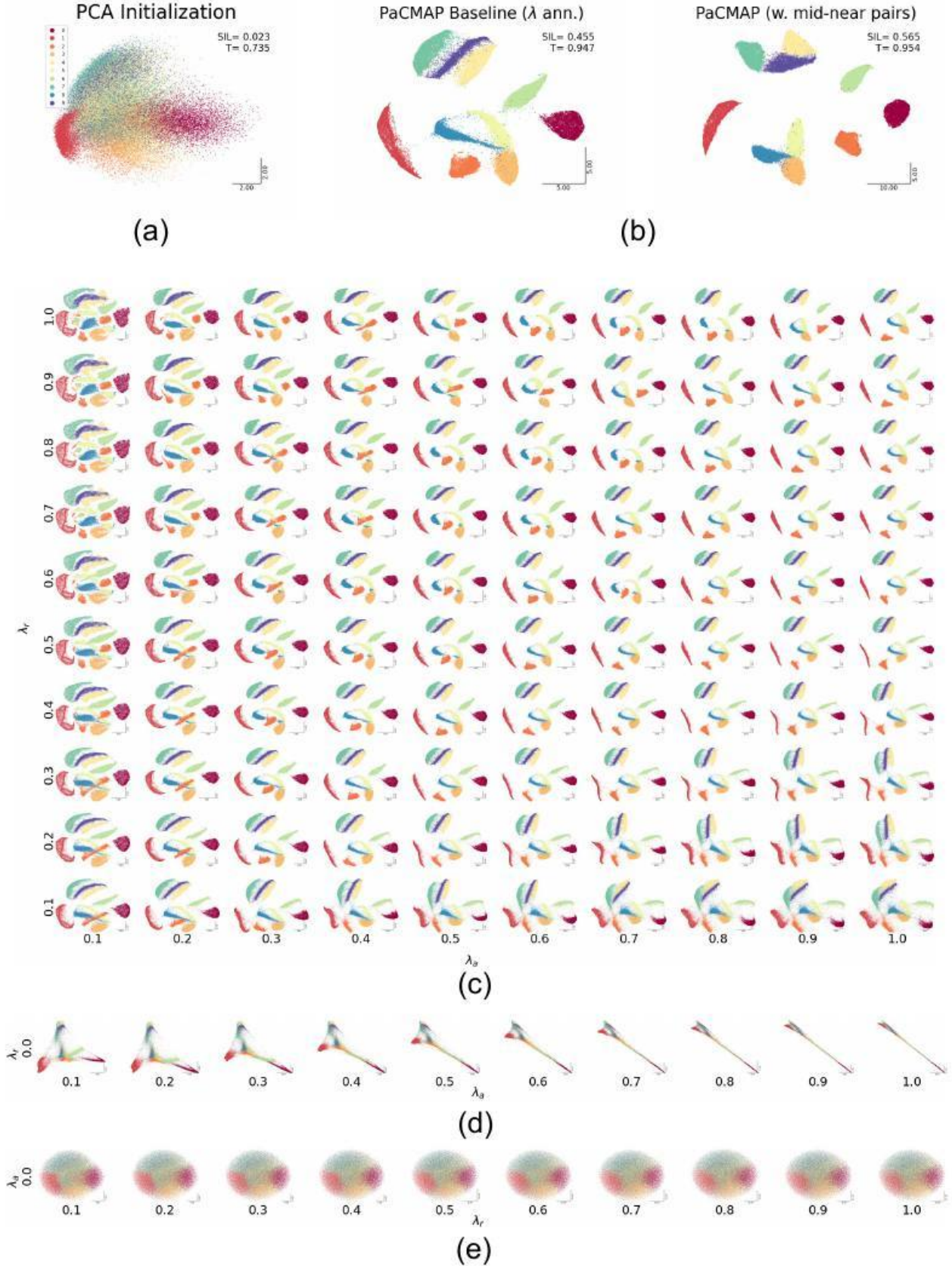


Figure 22: MNIST embeddings while varying λ_a and λ_r using PaCMAP's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Left: Baseline when λ is annealed from 1 (mid-near points are excluded to observe the interaction of attraction-repulsion alone), right: when mid-near points are considered. (c) The embeddings when λ_a and λ_r vary (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone cannot produce any cluster. (e) Similarly, when λ_a is set to 0 (no attraction), the repulsive force alone cannot produce any clusters.

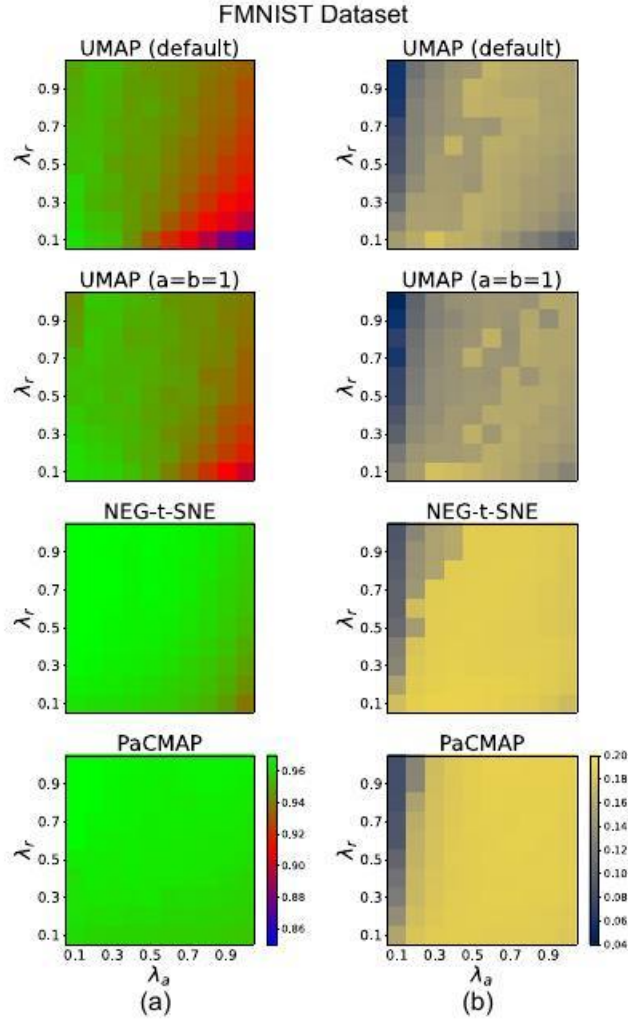


Figure 23: (a) Trustworthiness and (b) Silhouette score of different methods for the FMNIST dataset.

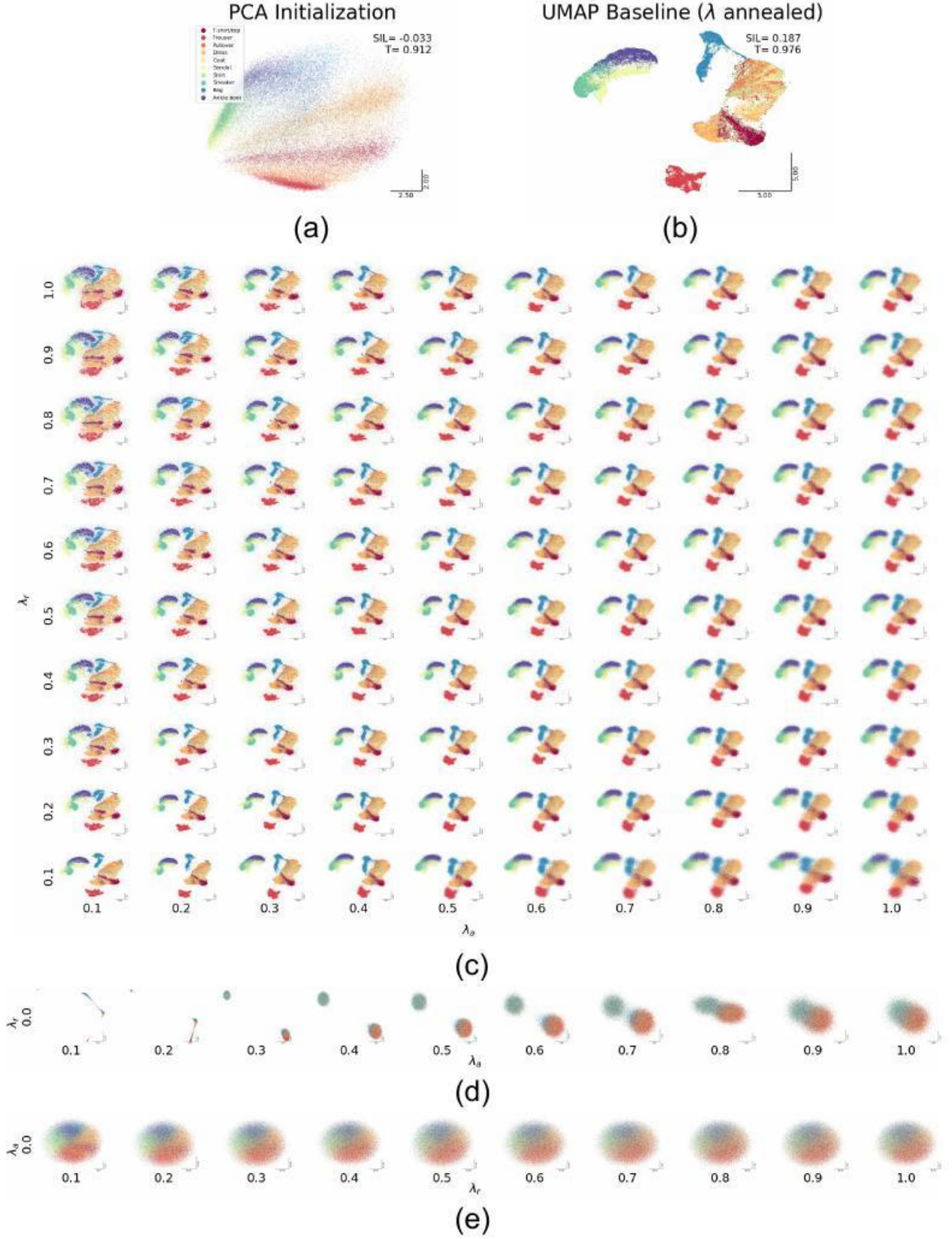


Figure 24: FMNIST embeddings while varying λ_a and λ_r using UMAP's attraction and repulsion shapes. (a) Initialization used for the embeddings. (b) Annealed-learning-rate baseline with λ starting from 1. (c) Embeddings obtained by varying λ_a and λ_r without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone does not produce clusters. (e) With $\lambda_a = 0$ (no attraction), repulsion alone also does not produce clusters.

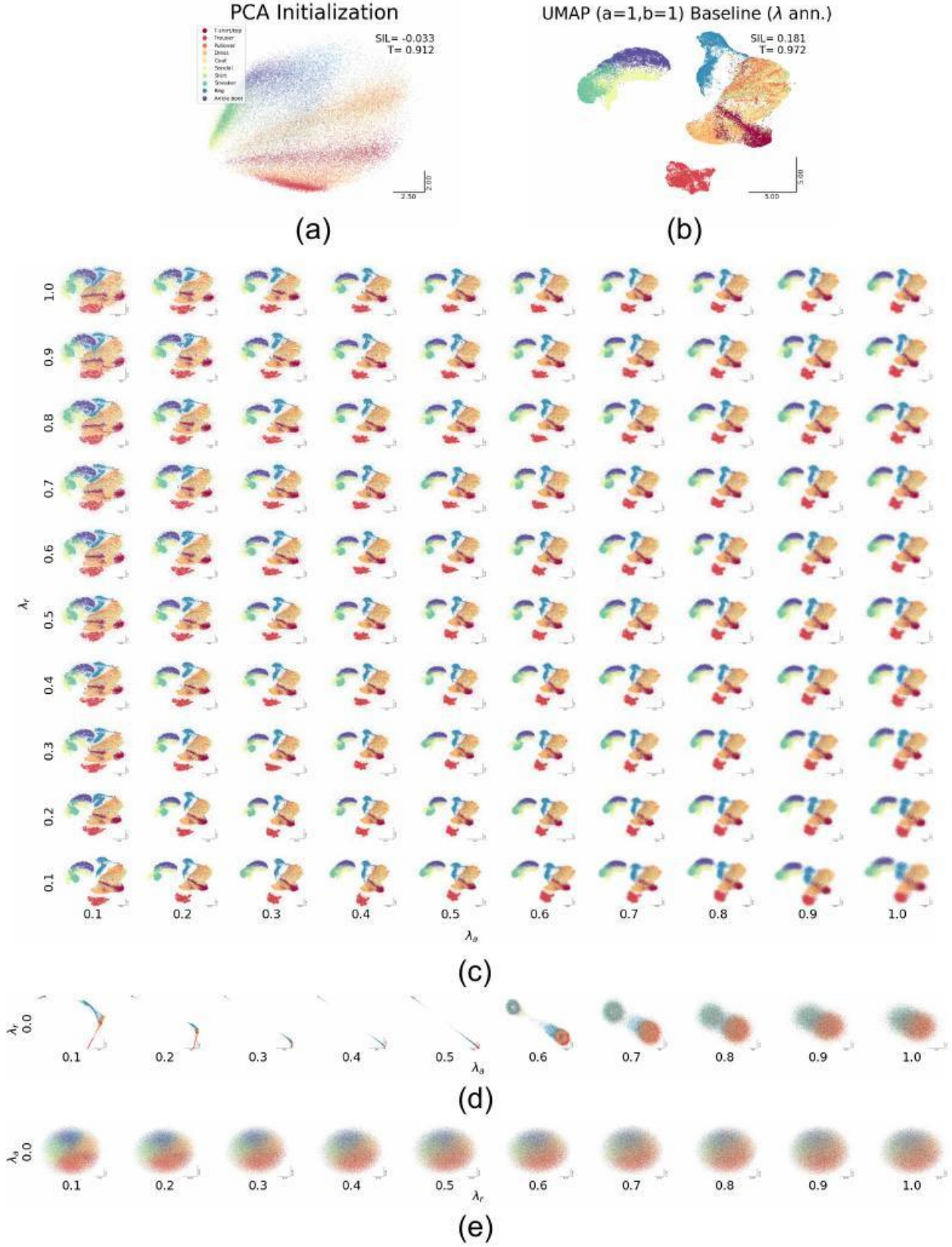


Figure 25: FMNIST embeddings while varying λ_a and λ_r using UMAP's attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Initialization used for the embeddings. (b) Annealed-learning-rate baseline with λ starting from 1. (c) Embeddings obtained by varying λ_a and λ_r without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone does not produce clusters. (e) With $\lambda_a = 0$ (no attraction), repulsion alone also does not produce clusters.

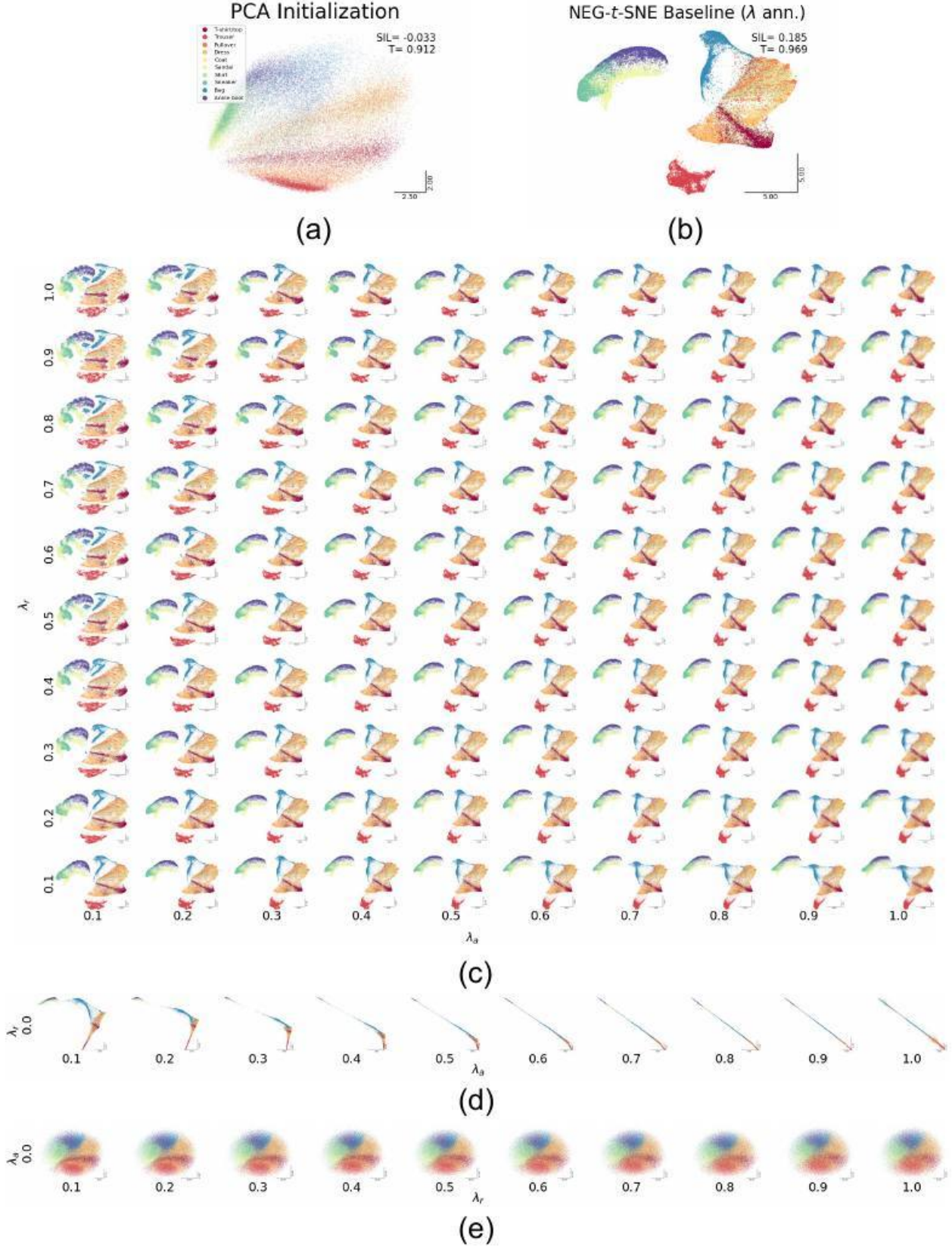


Figure 26: FMNIST embeddings while varying λ_a and λ_r using NEG-t-SNE's attraction and repulsion shapes. (a) Initialization used for the embeddings. (b) Annealed-learning-rate baseline with λ starting from 1. (c) Embeddings obtained by varying λ_a and λ_r without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone does not produce clusters. (e) With $\lambda_a = 0$ (no attraction), repulsion alone also does not produce clusters.

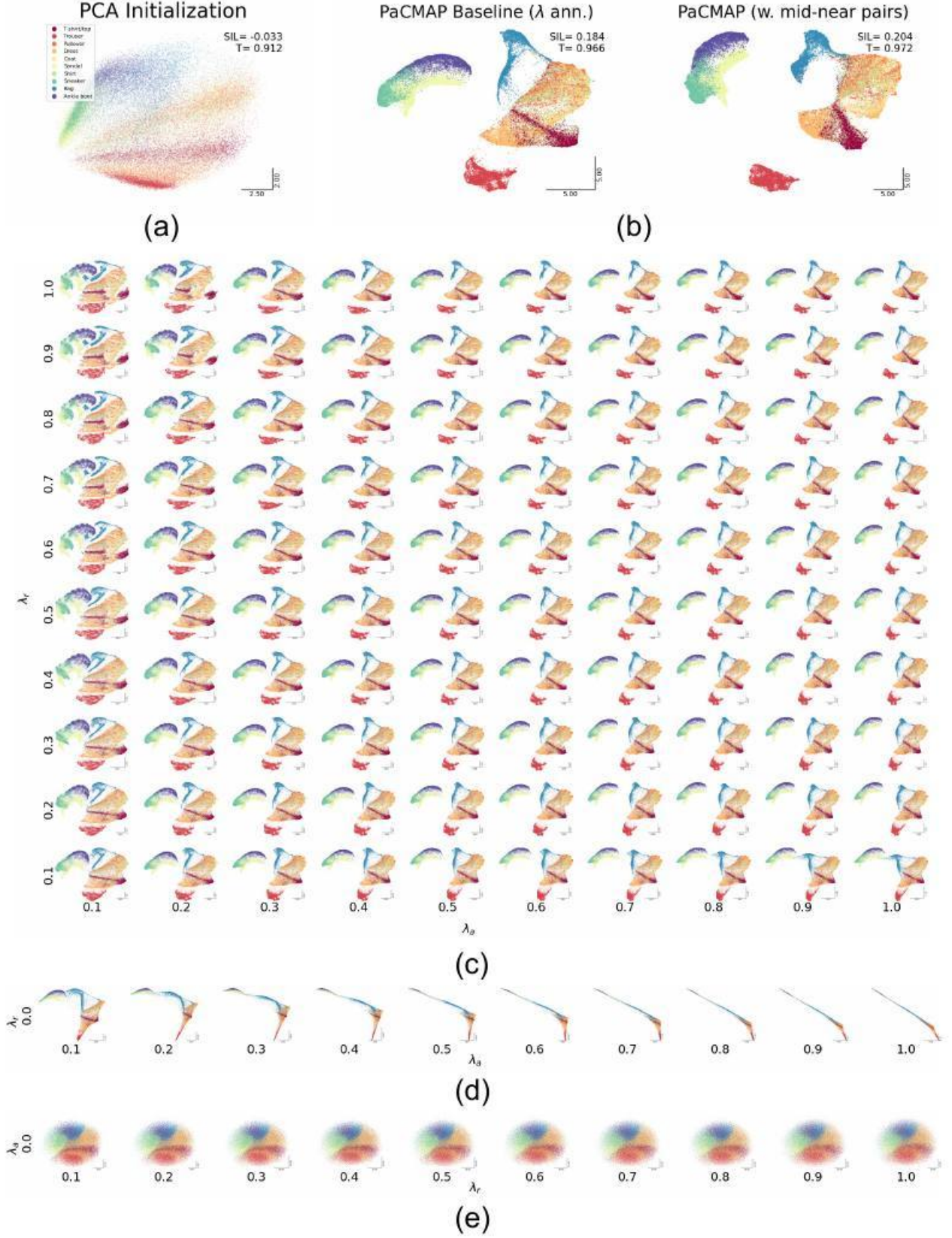


Figure 27: FMNIST embeddings while varying λ_a and λ_r using PaCMAP’s attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Left: Baseline when λ is annealed from 1 (mid-near points are excluded to observe the interaction of attraction-repulsion alone), right: when mid-near points are considered. (c) The embeddings when λ_a and λ_r vary (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone cannot produce any cluster. (e) Similarly, when λ_a is set to 0 (no attraction), the repulsive force alone cannot produce any clusters.

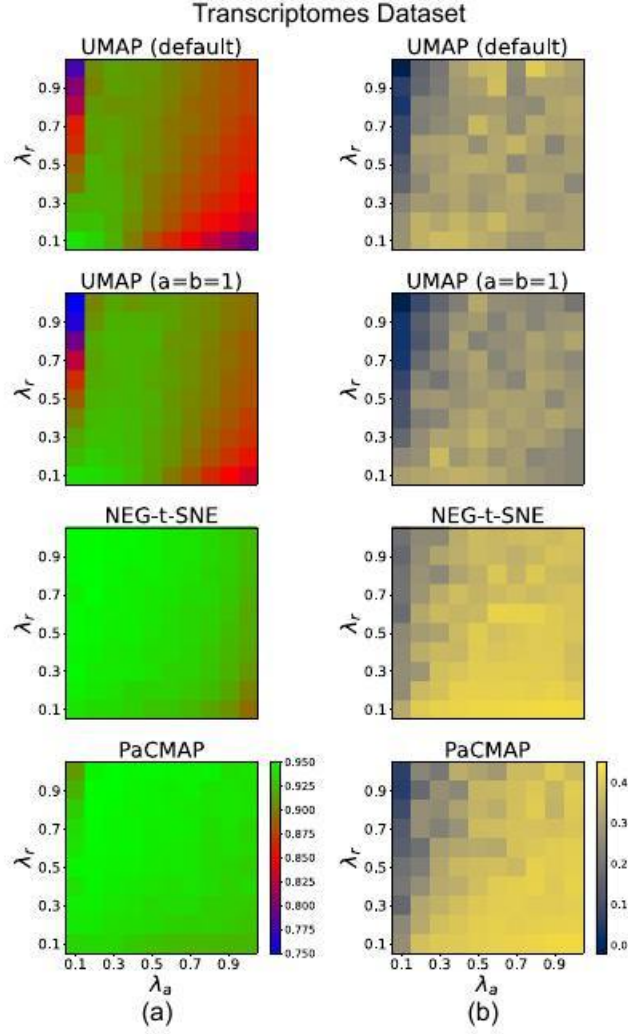


Figure 28: (a) Trustworthiness and (b) Silhouette score of different methods for the Single-cell transcriptomes dataset.

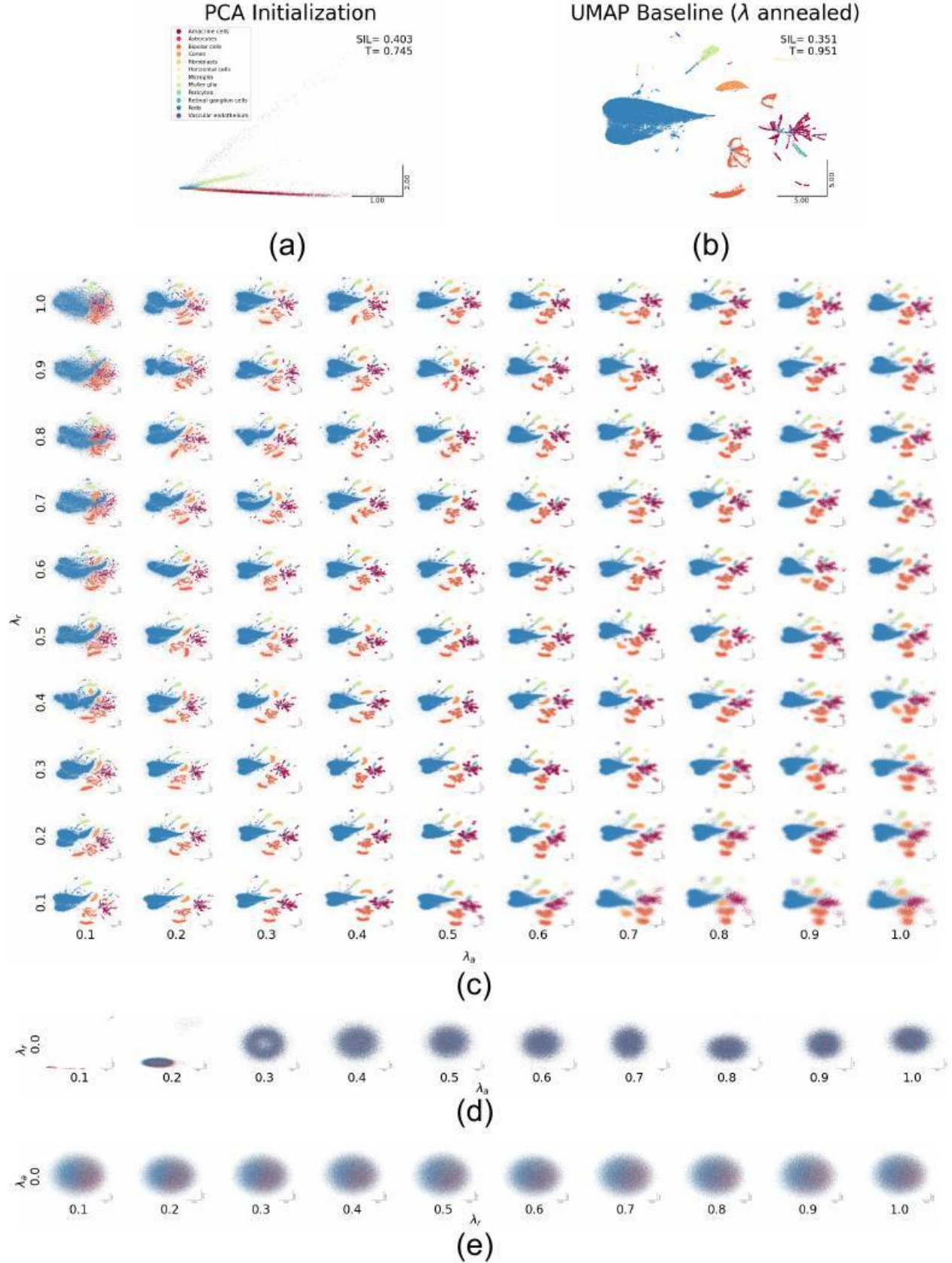


Figure 29: Single-cell transcriptomes embeddings while varying λ_a and λ_r using UMAP's attraction and repulsion shapes. (a) Initialization used for the embeddings. (b) Annealed-learning-rate baseline with λ starting from 1. (c) Embeddings obtained by varying λ_a and λ_r without annealing. (d) With $\lambda_r = 0$ (no repulsion), attraction alone does not produce clusters. (e) With $\lambda_a = 0$ (no attraction), repulsion alone also does not produce clusters.

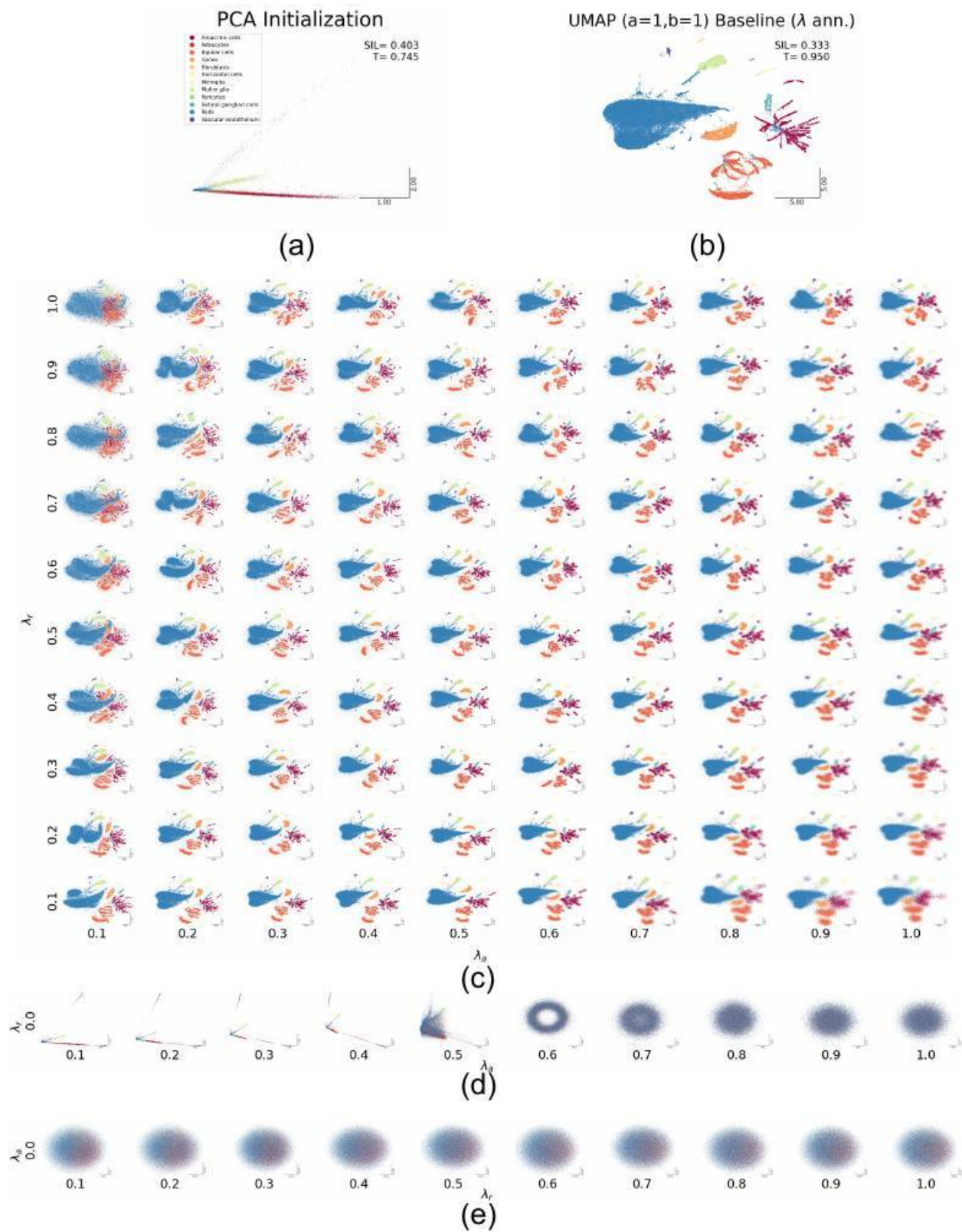


Figure 30: Single-cell transcriptomes embeddings while varying λ_a and λ_r using UMAP’s attraction and repulsion shapes (by setting $a = 1$ and $b = 1$). (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings when λ_a and λ_r vary (without any annealing). (d) When λ_r is set to 0 (no repulsion), attractive force alone cannot produce any cluster. (e) Similarly, when λ_a is set to 0 (no attraction), repulsive force alone cannot produce any clusters.

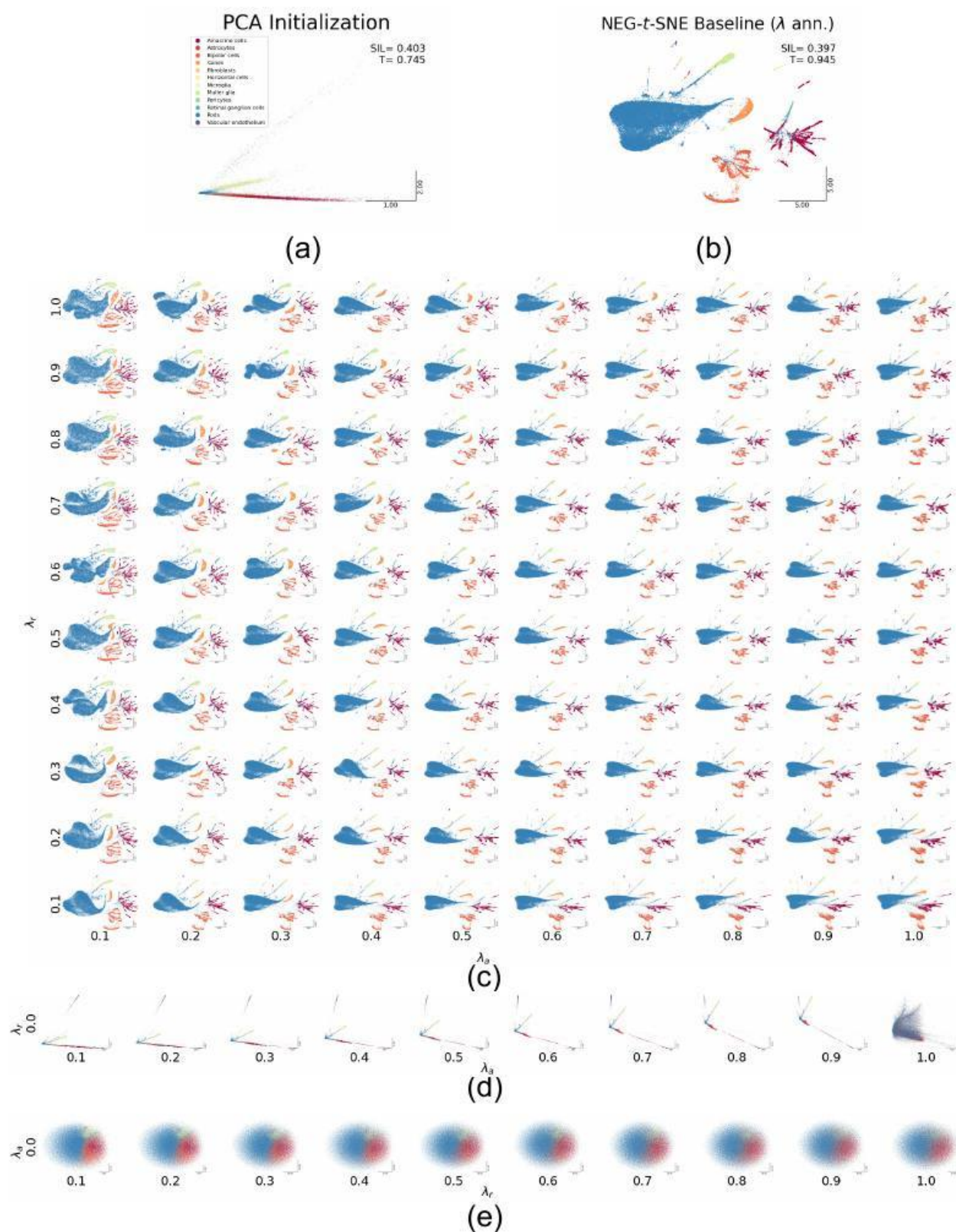


Figure 31: Single-cell transcriptomes embeddings while varying λ_a and λ_r using NEG-*t*-SNE’s attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Baseline when λ is annealed from 1. (c) The embeddings when λ_a and λ_r vary (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone cannot produce distinct clusters. (e) Similarly, when λ_a is set to 0 (no attraction), the repulsive force alone cannot produce any clusters.

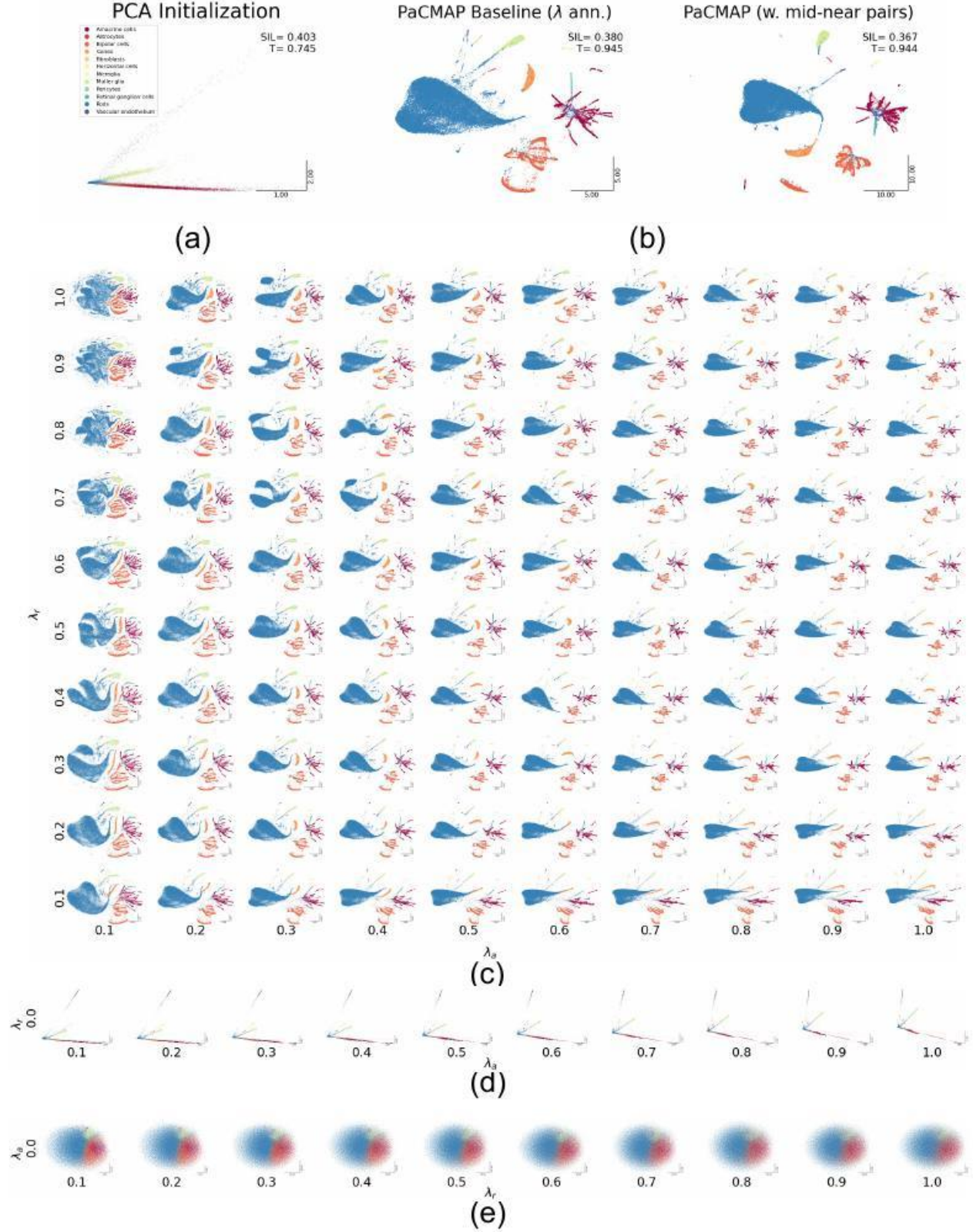


Figure 32: Single-cell transcriptomes embeddings while varying λ_a and λ_r using PaCMAP's attraction and repulsion shapes. (a) Initialization for the embeddings. (b) Left: Baseline when λ is annealed from 1 (mid-near points are excluded to observe the interaction of attraction-repulsion alone), right: when mid-near points are considered. (c) The embeddings when λ_a and λ_r vary (without any annealing). (d) When λ_r is set to 0 (no repulsion), the attractive force alone cannot produce any cluster. (e) Similarly, when λ_a is set to 0 (no attraction), the repulsive force alone cannot produce any clusters.

J Mix and Match Approach

Our analysis treats attraction and repulsion shapes as separable design objects. This suggests that shapes derived from different losses and algorithmic principles can be recombined while keeping the optimization protocol fixed. Figure 33 shows examples of this idea. We restrict attraction shapes to $[-1, 0]$ so that they satisfy Proposition 4.1 and are less sensitive to learning rates. We also use finite repulsion shapes to reduce the impact of very large repulsive forces at small distances. The learning rate is held constant ($\lambda = 1$), and the same UMAP optimization is used throughout, so differences arise from shape choices rather than learning-rate schedules or optimizers. In these MNIST experiments, UMAP’s attraction shape (with $f_a = 0.5f_a^U$) gives better clustering by silhouette score, whereas PaCMAP’s attraction shape gives better local structure by trustworthiness. The latter follows from PaCMAP’s attraction shape lying in $[-0.5, 0]$, which prevents flips during updates. Confining UMAP and NEG- t -SNE attraction shapes to $[-0.5, 0]$ (using $f_a = 0.25f_a^U$ and $f_a = 0.5f_a^N$, respectively) improves local structure at the cost of clustering performance (Fig. 34).

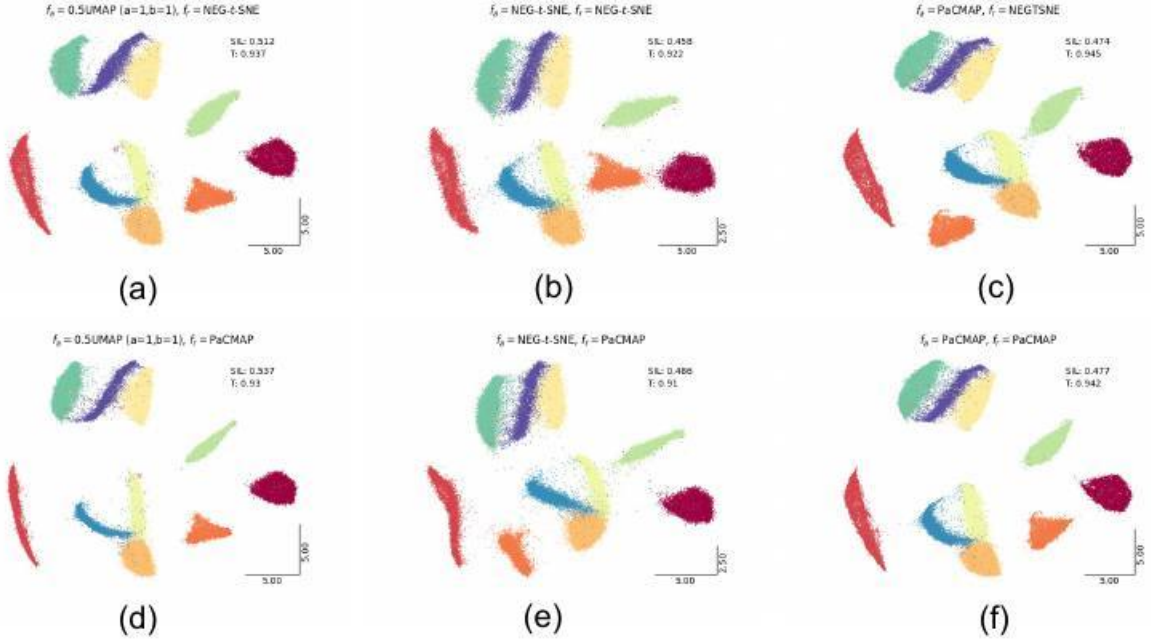


Figure 33: MNIST embeddings from mixed attraction and repulsion shapes at constant learning rate ($\lambda = 1$). Attraction shapes satisfy Proposition 4.1 and repulsion shapes are finite. Top row: UMAP, NEG- t -SNE, and PaCMAP attraction shapes paired with NEG- t -SNE repulsion. Bottom row: the same attraction shapes paired with PaCMAP repulsion.

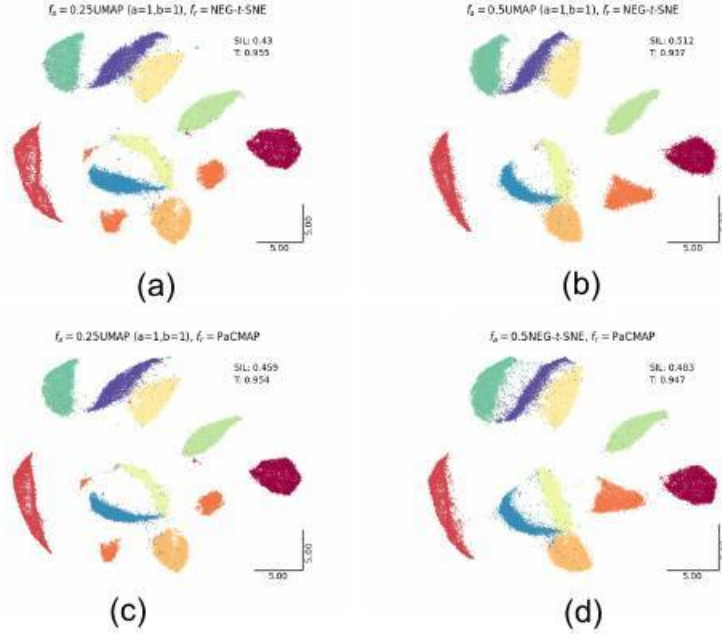


Figure 34: MNIST embeddings from mixed shapes at constant learning rate ($\lambda = 1$) with attraction confined to $[-0.5, 0]$ to prevent flips during attractive updates. Top row: UMAP and NEG- t -SNE attraction shapes paired with NEG- t -SNE repulsion. Bottom row: UMAP and NEG- t -SNE attraction shapes paired with PaCMAP repulsion.

K Implementation Details

For the shape analyses, we implemented a UMAP algorithm that allowed direct control over attraction and repulsion shapes. We used `numba` (Lam et al., 2015) to compute an exact nearest-neighbor graph, rather than an approximate one, with $k = 15$, and used `scikit-learn`'s (Pedregosa et al., 2011) PCA implementation for PCA initialization. This implementation produced and quantified the embeddings in Figs. 1, 2, 3, 4, 6, 8, 9 and 10. Unless otherwise stated, we used the same implementation when replacing UMAP's attraction and repulsion shapes with those of alternative methods (Figs. 18, 34). Trustworthiness and silhouette scores were computed with the corresponding `scikit-learn` functions.

The mappings in Figs. 2, 3, 9, 10 and 15 are aligned to the corresponding reference embedding, panel (a) in each figure. We perform Procrustes alignment by normalizing embeddings to zero mean and unit norm and then using SciPy's (Virtanen et al., 2020) `orthogonal_procrustes` method to obtain rotation and scaling parameters.

For the comparison with Neg- t -SNE in Fig. 11 we used the original contrastive-embedding framework implementation for both UMAP and Neg- t -SNE; implementation details will be released upon acceptance.

PaCMAP and LocalMAP embeddings in Figs. 15, 16, 22 (b) (right), 27 (b) (right), and 32 (b) (right) were generated with the official PaCMAP package; implementation details will be released upon acceptance.

Code for displaying the PubMed embeddings in Fig. 7 will be released upon acceptance.